

# Mutual influence of terms in a semi-empirical mass formula

Michael W. Kirson\*

*Department of Particle Physics, Weizmann Institute of Science, 76100 Rehovot, Israel*

Received 25 September 2007; received in revised form 27 October 2007; accepted 29 October 2007

Available online 1 November 2007

---

## Abstract

The semi-empirical mass formula of Bethe and von Weizsäcker is extended to include a number of additional terms. All possible combinations of extra terms are fitted in turn to the measured nuclear masses and the results analysed in order to reveal correlations and mutual influences between the various terms. The surface symmetry term is shown to be of critical importance in determining the Wigner and symmetry coefficients, while the pairing and global shell correction terms are largely independent of the other terms in the formula. The curvature term has a strong effect on the values of the other fit parameters. The interplay between different terms is found to be important. Some of the coefficients turn out to be much more poorly determined than would be suggested by standard fit errors alone.

© 2007 Elsevier B.V. All rights reserved.

PACS: 21.10.Dr

Keywords: Mass formula; Binding energy

---

## 1. Introduction

Semi-empirical mass formulas have been a staple of nuclear physics since shortly after the identification of the neutron as a constituent of the nucleus. The venerable formula of Bethe and von Weizsäcker, which was described in detail in “Bethe’s bible” [1], has been extended and modified frequently over the years. Proliferation of terms has sometimes threatened to overwhelm the original purpose—to obtain as simple as possible an understanding of the dependence

---

\* Tel.: +972 8 9342067; fax: +972 8 9344106.

E-mail address: [michael.kirson@weizmann.ac.il](mailto:michael.kirson@weizmann.ac.il).

of nuclear masses on neutron number  $N$  and proton number  $Z$ , using theoretical arguments to fix the form of the  $N$ ,  $Z$  dependence and measured masses to determine the coefficients of the various terms.

The recent dramatic expansion of experimental nuclear physics facilities, allowing the measurement of nuclear properties very far from the long-familiar valley of stability, has triggered, amongst others, a renewed interest in nuclear astrophysics. Although the masses of many very exotic nuclei are now amenable to measurement, many more nuclei of great interest in astrophysics remain beyond the capabilities of modern mass measurements. This has conferred increased urgency on the development of reliable extrapolations of measured masses, one of the classic applications of mass formulas. Microscopic mass formulas are probably the preferred instrument for such extrapolations, based as they are on well-established theory and having much smaller root-mean-squared deviations from measured masses than semi-empirical formulas can achieve. (See [2] for an impressive example. It may also be noted that a rather similar, but more restricted, approach [3] to that adopted here, but including calculated microscopic shell corrections, achieves rms deviations smaller by almost a factor of three than the best results obtained in the present work.) However, there is significance in the numerical values of the coefficients of the semi-empirical mass formula, influencing, for instance, the deduction of the binding energy per nucleon of infinite nuclear matter, or the symmetry energy and surface tension of nuclear matter. These are not easily accessed by the microscopic mass formulas.

In order to understand and appreciate the relevance and reliability of the semi-empirical mass formula, it is necessary to study not only the effect of adding specific terms to the original Bethe–Weizsäcker formula but also the interplay between the different terms. Although there is a rich history of proposing additional terms for inclusion in the formula, there has been much less attention paid to the way the different terms influence one another. It is the purpose of the present investigation to rectify this oversight.

The most recent comprehensive database of nuclear masses is the impressive atomic mass evaluation generally referred to as AME2003 [4] (very unfortunately, probably the last of its kind). This tabulation constitutes the raw material for this work. The relevant input is the list of measured binding energies of nuclei, obtained by multiplying the tabulated binding energy per nucleon by the mass number  $A$ . Only measured nuclei are included—the many masses extrapolated from systematics and marked with the symbol # in the error column are not taken into account here.

Fit parameters, the coefficients of the various terms in the mass formula, are determined by linear regression (least squares) and are quoted together with their standard errors, shown in parentheses and referring to the least significant digits of the coefficients. (For instance, 123.45(6) means  $123.45 \pm 0.06$ .) Goodness of fit will be measured by the rms deviation of the fit from the measured binding energies, defined by

$$D_{\text{rms}} = \sqrt{\frac{\sum_i (M_i - F_i)^2}{n - f}}, \quad (1.1)$$

where  $i$  labels the data points,  $M_i$  is the measured binding energy,  $F_i$  is the value predicted by the fit,  $n$  is the total number of data points (2225, for the AME2003 tabulation) and  $f$  is the number of free parameters in the fit. The quantity  $D_{\text{rms}}^{\text{BW}}$ , the rms deviation of the baseline Bethe–Weizsäcker formula, will be used as a reference value, with the effect of changes in the mass formula being measured by

$$\Delta_{\text{rms}} = D_{\text{rms}} - D_{\text{rms}}^{\text{BW}}. \quad (1.2)$$

The basis for the classic Bethe–Weizsäcker mass formula is sketched in Section 2 of this paper, and the result of fitting it to the AME2003 database is given. Section 3 describes the additional terms to be added to the formula, six in all, with the arguments for their inclusion. The results of fitting all possible combinations of the extra terms are given in Section 4, together with discussion of the resulting rms deviations and parameter values, including mutual influences and correlations. Section 5 is devoted to alternative versions of some of the extra terms. The paper is summarized in Section 6, which also presents a discussion of the principal results.

## 2. Bethe–Weizsäcker mass formula

The total binding energy of the nucleus depends principally on the mass number  $A$ , as is evident from Fig. 1, where a plot of binding energy against mass number for all measured nuclei in the 2003 mass evaluation shows a very narrow band of data. The width of the band ranges up to 66 MeV, with a mean value of 28 MeV (only 1.4% of the range of values of  $B$ ) and a standard deviation of 14 MeV. The total binding energy is almost proportional to  $A$ , with a slow falloff at large  $A$ . Of the 2225 nuclei, only 82, all of them having  $A < 40$  (and all but 7 having  $A < 25$ ), have a binding energy per nucleon differing by more than 10% from the mean value of 8.1 MeV. A single-term mass formula  $B = 7.95(1) A$  MeV has a root-mean-squared deviation from the measured masses of 48.416 MeV, a rather good fit to a quantity varying from 2 MeV to 1943 MeV.

This leading linear behaviour of the binding energy as a function of the number of nucleons is a manifestation of the saturation of nuclear forces. Another manifestation of saturation is the dependence of the nuclear volume on mass number. Nuclei have diffuse surfaces, so the notion of volume is somewhat fuzzy, but an indicator is provided by the measured root-mean-squared (rms) radius,  $\langle r^2 \rangle^{1/2}$ , which sets the linear scale of nuclear size. The cube of this quantity,  $\langle r^2 \rangle^{3/2}$ , should track the nuclear volume reasonably well. It is plotted against  $A$  in Fig. 2, using the rms radii tabulated by Angeli [5], and also reveals near proportionality to, and dominant dependence on,  $A$ . The best fit to 798 nuclei is  $\langle r^2 \rangle^{3/2} = 0.842(1) A$  fm<sup>3</sup>, which implies  $\langle r^2 \rangle^{1/2} = 0.944(1) A^{1/3}$  fm. The volume of a uniform sphere having this cubed rms radius is 7.59(1) fm<sup>3</sup>, corresponding to a number density of 0.132 nucleons per fm<sup>3</sup>.

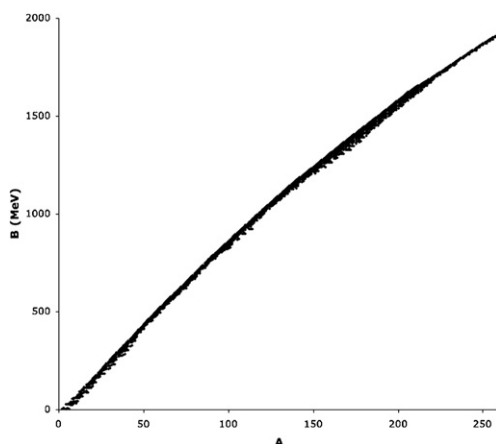


Fig. 1. Dependence of total binding energy  $B$  (in MeV) on mass number  $A$ .

Following Bethe [1], the proportionality of both the total binding energy and the nuclear volume to the number of nucleons can be regarded as showing that the binding energy is, to a first approximation, proportional to the nuclear volume. By analogy with the properties of a liquid drop, it is to be expected that the binding will then fall off in the surface of the nucleus, by an amount proportional to the surface area. This surface tension effect is incorporated in a two-term mass formula,  $B = \alpha_v A + \alpha_s A^{2/3}$ , where the volume coefficient  $\alpha_v$  is expected to be positive, the surface coefficient  $\alpha_s$  negative. A fit to the measured masses produces  $\alpha_v = 5.68(3)$  MeV,  $\alpha_s = 12.62(19)$  MeV, with rms deviation reduced by almost a factor of 2, to 28.269 MeV. Both the counter-intuitive sign of  $\alpha_s$  and the systematic nature of the deviations of the fit from the data indicate significant missing physics at this level.

An obvious missing feature in the two-term formula is the effect of the repulsive Coulomb interaction between protons. On the basis of simple electrostatics and dimensional arguments, the total Coulomb energy of the nucleus is expected to be proportional to the square of its charge and inversely proportional to its length scale,  $E_C \propto Z^2 e^2 / A^{1/3}$ , where  $e^2 = 1.44$  MeV fm in convenient units. It may be noted that any classical charge distribution, of total charge  $Q$ , depending on only a single length scale, has Coulomb energy  $E_C = k Q^2 / \langle r^2 \rangle^{1/2}$ , with  $k$  a constant. The value of  $k$  turns out to be close to  $1/2$  for a number of distributions (linear, uniform, parabolic, exponential, Gaussian), suggesting a nuclear Coulomb energy  $E_C \approx 0.7 Z^2 / A^{1/3}$  MeV and a three-term mass formula  $B = \alpha_v A + \alpha_s A^{2/3} + \alpha_C Z^2 / A^{1/3}$ . Here, again, it is anticipated that  $\alpha_v$  will be positive,  $\alpha_s$  negative and the Coulomb coefficient  $\alpha_C$  also negative. A fit produces  $\alpha_v = 7.67(9)$  MeV,  $\alpha_s = 6.80(31)$  MeV,  $\alpha_C = -0.174(8)$  MeV, again with unintuitive fit coefficients and strong systematic deviations, and with an rms deviation only slightly reduced, to 25.499 MeV.

The final missing link leading to the Bethe–Weizsäcker mass formula is the symmetry term, producing a rough balance between the number of neutrons  $N$  and number of protons  $Z$  among the  $A = N + Z$  nucleons. In the absence of this term, the previous formula would suggest maximum binding for a nucleus consisting only of neutrons. The facts that the only bound two-nucleon system is the deuteron (one neutron, one proton) and that no bound pure multi-neutron or pure multi-proton nuclei have been detected indicate a mechanism favouring  $N \approx Z$ . The nuclei in the mass table have an  $N/Z$  ratio converging quite tightly on  $N/Z \approx 1.5$  for medium

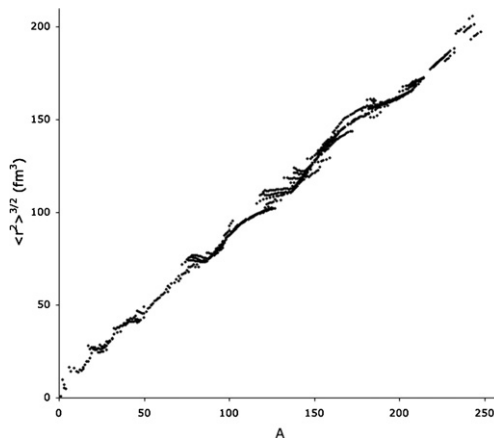


Fig. 2. Dependence of volume indicator  $\langle r^2 \rangle^{3/2}$  (in fm<sup>3</sup>) on mass number  $A$ .

to large  $A$ , so that  $Z \approx 0.4A$  and the Coulomb term can account qualitatively for the steady decrease in binding energy per nucleon at large  $A$ . There are two contributing factors to the symmetry energy—a kinetic energy term and a potential energy term [6].

The kinetic energy of the nucleus can be estimated in a Fermi gas model. In a cubic box of volume  $\Omega$  with infinitely repulsive walls, fermion single-particle states are filled up to the Fermi momentum  $k_F$ , which is determined by the number density  $\rho = A/\Omega$  via  $k_F = (6\pi^2\rho/g)^{1/3}$ , where  $g$  is the degeneracy of the single-particle level ( $g = 2$  for spin-1/2 fermions,  $g = 4$  for spin-1/2, isospin-1/2 nucleons). The total kinetic energy is then  $E_K = 3A\hbar^2 k_F^2/10m$ , for nucleons of mass  $m$ , and is proportional to  $A^{5/3}$ . If  $N \neq Z$ , but neutrons and protons occupy the same volume  $\Omega$ , the total kinetic energy is  $E_K \propto N^{5/3} + Z^{5/3}$ . In terms of  $A = N + Z$  and  $\Delta = N - Z$ , expanding to lowest non-vanishing order in the small quantity  $\Delta/A$ , this is proportional to  $A^{5/3}[1 + \frac{5}{9}(\Delta/A)^2]$ , indicating an increase in total kinetic energy, proportional to  $(N - Z)^2/A$ , when  $N$  differs from  $Z$ . A similar argument produces the same result if an isotropic harmonic oscillator picture is used instead of the infinite cubic well.

A qualitative estimate of the potential energy contribution is obtained by considering the isovector part of the nucleon–nucleon interaction,  $\frac{1}{2} \sum_{i,j} v_{ij} \vec{\tau}^{(i)} \cdot \vec{\tau}^{(j)}$ , where  $\vec{\tau} = 2\vec{t}$  is the isospin operator of a nucleon. (The isoscalar part of the interaction does not distinguish between protons and neutrons.) In a state with a fixed number of neutrons and protons, the expectation value of this interaction is approximately  $\langle v \rangle \sum_{i,j} \tau_z^{(i)} \tau_z^{(j)}$ . Since the average matrix element of a short-range interaction  $v_{ij}$  is inversely proportional to the volume occupied by the state of interest, and in nuclei the volume is proportional to  $A$ , while  $\sum_i \tau_z^{(i)} = Z - N$ , this becomes proportional to  $(N - Z)^2/A$ , the same form as for the kinetic energy. Taken together, the two contributions lead to the final form of the Bethe–Weizsäcker mass formula,

$$B = \alpha_v A + \alpha_s A^{2/3} + \alpha_C Z^2/A^{1/3} + \alpha_t (N - Z)^2/A, \quad (2.1)$$

where  $\alpha_v$  is expected to be positive, the other three  $\alpha$ 's negative. This formula will be referred to from now on as BW. (Note the subscript  $t$  on the coefficient of the symmetry term, recalling its connection to the isospin  $T$ .)

It should be noted that more sophisticated treatments of the nuclear interaction (as in the seniority shell model or the Wigner supermultiplet theory [7]) produce symmetry energies of the form  $T(T + 1)$  or  $T(T + 4)$ , respectively, where the nuclear isospin  $T = |N - Z|/2$ . This corresponds roughly to the replacement, in the above treatment, of  $\sum_{i,j} \vec{\tau}^{(i)} \cdot \vec{\tau}^{(j)}$  by  $4\vec{T} \cdot \vec{T}$ , with  $\vec{T} \cdot \vec{T}$  having eigenvalue  $T(T + 1)$ . This modification will be considered further below.

The BW mass formula, Eq. (2.1), provides the baseline fit, relative to which all refinements can be judged. The fit to the 2003 mass table produces

$$\begin{aligned} \alpha_v &= 15.36(3) \text{ MeV}, \\ \alpha_s &= -16.42(8) \text{ MeV}, \\ \alpha_C &= -0.691(2) \text{ MeV}, \\ \alpha_t &= -22.53(7) \text{ MeV}, \\ D_{\text{rms}}^{\text{BW}} &= 3.536 \text{ MeV}, \end{aligned} \quad (2.2)$$

an order of magnitude improvement in the rms deviation over the previous partial fits. It is noteworthy that the signs of the fit coefficients now conform to expectations, with the Coulomb coefficient, in particular, astonishingly close to its estimated value. Comparison of the symmetry coefficient with 5/9 of the kinetic energy per nucleon in the Fermi gas model suggests that

the potential energy contribution to this coefficient is comparable to, but larger than, the kinetic energy contribution.

### 3. Additional mass formula terms

Over the years, many suggestions have been made for supplementing the semi-empirical BW mass formula by additional terms designed to address apparently omitted physics. Perhaps the most extensive early expansion of the mass formula was via the droplet model [8], which introduced many more terms, frequently constrained by theoretical considerations. A number of such terms have been selected for further discussion here, with their effects on the quality of the fit to the measured masses being presented later.

#### 3.1. Exchange Coulomb term

It has been remarked that the Coulomb term in the BW mass formula is in remarkable agreement with the a priori estimate. This can be further tested by considering mirror nuclei, pairs of nuclei with same value of  $A$ , but with  $N_1 = Z_2$  and  $Z_1 = N_2$ , i.e. with neutrons and protons interchanged. Given charge independence of the nuclear force, the binding energies of mirror pairs can differ only in their Coulomb energies. (For instance, all terms in the BW mass formula are symmetric in  $N$  and  $Z$ , except the Coulomb term.) The difference in binding energy between two mirror nuclei is thus  $\Delta B = \alpha_C(N^2 - Z^2)/A^{1/3} = \alpha_C \Delta Z A^{2/3}$ , where  $\Delta Z$  is the difference in proton number between the two mirror nuclei. The quantity  $\Delta B/\Delta Z$  should be linear in  $A^{2/3}$ , with coefficient of proportionality equal to  $\alpha_C$ .

In Fig. 3 the binding energy differences, scaled by charge difference  $\Delta Z$ , of the 88 measured mirror nucleus pairs in the 2003 mass table are plotted against  $A^{2/3}$  and are seen to lie on a straight line. The value of  $\Delta Z$  ranges from 1 (32 cases) to 5 (1 case). The best fit straight line has a slope of 0.716(4) MeV, very close to the value of the Coulomb coefficient  $\alpha_C$ , but the intercept is  $-1.051(39)$  MeV, a sizeable distance from the expected value of zero. The implication is that a physical effect has been overlooked. It might be argued that, since mirror pairs are limited to  $A \leq 75$ , this is a small- $A$  artefact. This may be checked by extending the analysis to isobaric analog states, which belong to the same isospin multiplet as neighbouring ground states and would have exactly the same energy if isospin were strictly conserved. The relative excitation

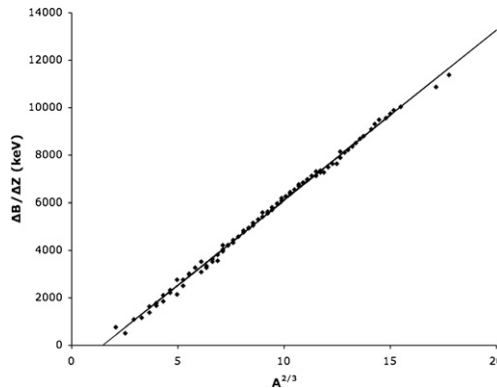


Fig. 3. Scaled mirror nucleus binding energy differences vs.  $A^{2/3}$  and linear fit.

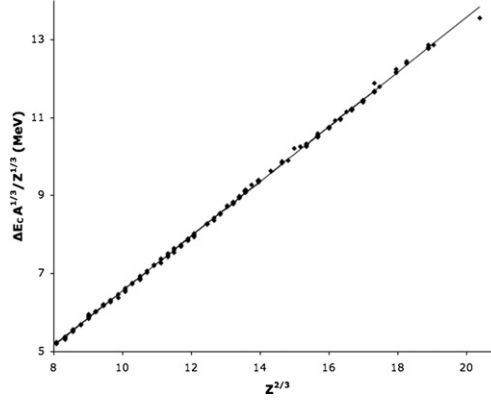


Fig. 4. Scaled Coulomb displacement energies vs.  $Z^{2/3}$  and linear fit.

energy of the analog state is again a measure of Coulomb energy differences,  $\Delta E_C = \alpha_C [(Z + 1)^2 - Z^2] / A^{1/3} \approx 2\alpha_C Z / A^{1/3}$ . The quantity  $A^{1/3} \Delta E_C / 2Z^{1/3}$  should be proportional to  $Z^{2/3}$ , with coefficient of proportionality  $\alpha_C$ . (The reason for this particular choice will become clear shortly.) Using 194 Coulomb displacement energies out of those tabulated in Ref. [9], Fig. 4 is obtained. (The entries selected from Ref. [9] have  $A \geq 50$ , involve analog states of ground states, have measurement errors quoted and do not include mirror nucleus pairs.) The clean linear behaviour persists to  $A = 238$ , the best fit slope being  $0.702(1)$  MeV, close to the mirror nucleus value, with an intercept of  $-0.453(15)$  MeV, again far from zero.

The missing term responsible for the non-zero intercepts is very likely a quantal correction to the classical Coulomb energy formula, namely the exchange term in the expression  $E_C = \langle \Psi | V_C | \Psi \rangle = \sum_{a,b} \langle ab | e^2 / r_{12} | ab - ba \rangle$ , where the double sum is over all proton single-particle states in the Fermi sea. When evaluated in the Fermi gas model [10], the exchange term contributes an amount proportional to  $Z^2 / \Omega k_F^2$ . Since  $k_F \sim (Z / \Omega)^{1/3}$  and  $\Omega \sim A$ , the implied structure of the Coulomb exchange term is  $\alpha_{xC} Z^{4/3} / A^{1/3}$ . Including this term changes the mirror nucleus Coulomb energy difference to  $\Delta B / \Delta Z = \alpha_C A^{2/3} + (2^{5/3} / 3) \alpha_{xC}$ , to a good approximation, and the analog state Coulomb displacement energy to  $A^{1/3} \Delta E_C / 2Z^{1/3} = \alpha_C Z^{2/3} + 2\alpha_{xC} / 3$ , to the same approximation. (This explains the form of the plot in Fig. 4.) The non-zero intercepts are obtained naturally, the mirror nucleus fit suggesting  $\alpha_{xC} = -0.993(37)$  MeV and the Coulomb displacement energy fit suggesting  $\alpha_{xC} = -0.680(23)$  MeV. The estimates are of the same order of magnitude, but  $\alpha_{xC}$  is clearly less well determined than  $\alpha_C$ . A combined fit of mirror nucleus binding energy differences and Coulomb displacement energies produces the parameter values  $\alpha_C = -0.717(1)$  MeV and  $\alpha_{xC} = 0.928(19)$  MeV.

### 3.2. Wigner term

In Section 2, reference was made to the  $T(T + 1)$  or  $T(T + 4)$  form of isospin dependence arising in the seniority or supermultiplet models of ground-state energies. In almost all nuclei, the ground state has isospin  $T = |T_z| = |N - Z|/2$ , the exceptions being about a dozen odd-odd  $N = Z$  nuclei with ground-state spins  $0^+$  and isospin  $T = 1$ , rather than  $T = |N - Z|/2 = 0$ . The implication of such a form is that, in addition to the regular symmetry energy, with its  $(N - Z)^2 / A$  dependence, there should be a term linear in  $T$ , i.e. proportional to  $|N - Z|$ . This is a non-analytic  $N, Z$  dependence, which can produce cusps in the systematics of nuclear masses.

The extra term linear in  $|N - Z|$  is generally referred to as the Wigner term because of its appearance in the supermultiplet theory introduced by Wigner [11]. Since it has been argued that the symmetry energy arises from both kinetic and potential energy contributions, while presumably only the latter has a Wigner term, it is to be expected that the mass formula will depend on isospin in the form  $T(T + a)$ , where  $a$  is to be determined but is expected to be positive. The Wigner term therefore introduces into the mass formula a term of the form  $\alpha_W |N - Z|/A$ , where  $\alpha_W$  is expected to be of the same order of magnitude and same sign as  $\alpha_t$ .

It is of interest that Myers and Swiatecki originally argued for a rather different form of the Wigner term [12]. They suggested that its effect should be greatest for  $N = Z$  and should fall off with increasing asymmetry, advocating an exponential form  $\alpha'_W \exp(-\beta_W |N - Z|/A)$ . This suggestion will be revisited in a later section.

### 3.3. Surface symmetry term

It has been forcefully argued [13] that it is inconsistent to include a symmetry correction to the volume energy but not to the surface energy (or alternatively, to include a surface correction to the volume energy but not to the symmetry energy). Regarding the volume plus symmetry terms as  $\alpha_v A(1 + \beta_t I^2)$ , where  $\beta_t = \alpha_t/\alpha_v$  and  $I = (N - Z)/A$ , it seems reasonable to replace the surface term by  $\alpha_s A^{2/3}(1 + \beta_{st} I^2)$ , with  $\beta_{st} = \alpha_{st}/\alpha_s$ . Although this is not the form preferred in Ref. [13], which argues on thermodynamic grounds for a non-linear surface symmetry term, and although the expansion in series of that non-linear surface symmetry term does not converge rapidly enough to justify the present linear form as a lowest-order estimate, the linear term will be used here, both for convenience and because of its intuitive “derivation”. (In practice, the linear form turns out to be marginally superior to the non-linear form in the fits to be reported below.)

The surface symmetry term will be taken to have the form  $\alpha_{st}(N - Z)^2/A^{4/3}$ .

### 3.4. Pairing term

It is instructive to plot the residuals of the best fit to the data with the BW mass formula. As seen in Fig. 5, there is clear systematic behaviour. When plotted against the neutron number  $N$ ,

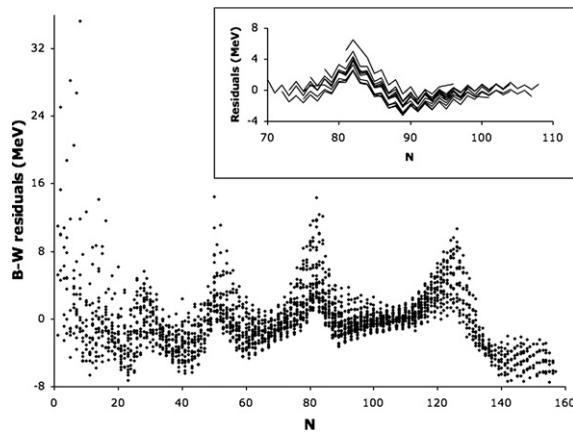


Fig. 5. Residuals of the BW fit to the mass data vs.  $N$ . The inset shows an expanded view of the segment of the plot for the elements Nd through Yb, with isotopes of a given element connected by lines.



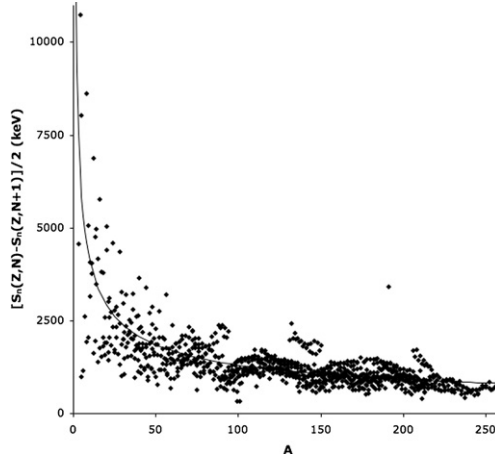


Fig. 6. Neutron pairing energy, as defined in Eq. (3.1), as a function of  $A$ . The curve is a  $1/\sqrt{A}$  fit to the data.

there is an evident parabolic trend between the magic numbers, which show up as cusps. In addition, when blown up in scale, chains of isotopes display a marked odd–even staggering, odd- $N$  isotopes of a given element being noticeably less bound than their neighbouring even- $N$  isotopes. (Similar behaviour is found when plotting the residuals against proton number  $Z$ .) This latter effect has been attributed to pairing of like nucleons and calls for an additional term in the mass formula to reflect it.

A quantitative measure of the pairing effect for neutrons is obtained by taking the difference between the binding energy of an even- $N$  isotope of a given element and the average binding energy of its two odd- $N$  neighbours. This difference is equal to one-half of the difference between even- $N$  and odd- $N$  single-neutron separation energies,

$$\begin{aligned}\Delta E_p(Z, N) &= B(Z, N) - [B(Z, N-1) + B(Z, N+1)]/2 \\ &= [S_n(Z, N) - S_n(Z, N+1)]/2.\end{aligned}\quad (3.1)$$

This quantity is shown as a function of  $A$  in Fig. 6. A power law gives a reasonable description of the general trend, the best fit being  $\Delta E_p(Z, N) = 9.01A^{-0.43}$  MeV. Similar behaviour is demonstrated by an analogous proton pairing energy. It would clearly be acceptable to adopt a  $1/\sqrt{A}$  form for the pairing term, as shown by the curve  $13/\sqrt{A}$  MeV in the figure.

An alternative presentation of the pairing effect is based on the observation that the binding energy of the nucleus, as given by the BW mass formula, is quadratic in  $Z$  for fixed  $A$ . For each  $A$ , the binding energy should lie on a parabola as a function of  $Z$ . For even values of  $A$ , the expected mass parabola is found to split into two roughly parallel parabolas, one for even–even and the other for odd–odd nuclei, the former lying higher. For odd  $A$ , a reasonably clean single mass parabola is seen. The resulting analysis is exemplified by Fig. 7, for  $A = 160$ , which shows the binding energies for even–even and odd–odd nuclei, together with parabolas fitted to the data points. In addition, an interpolated odd-mass parabola is displayed, through the average of the  $A = 159$  and  $A = 161$  binding energy at each  $Z$  value. It is apparent that the even–even nuclei are more bound than the odd- $A$  average, while the odd–odd nuclei are less bound than the odd- $A$  average. (In this particular case, the even–even nuclei are more bound by an average of 1.20 MeV, the odd–odd nuclei less bound by 0.55 MeV, corresponding to  $15/\sqrt{A}$  and  $7/\sqrt{A}$  MeV, respec-

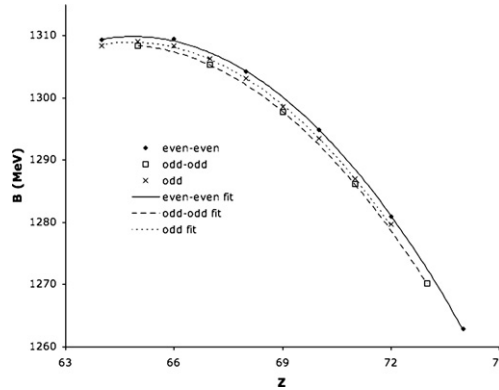


Fig. 7. Mass parabolas for  $A = 160$ . Even–even nuclei are represented by black diamonds, odd–odd nuclei by white squares and odd nuclei by crosses. The curves are parabolas fitted to the data points, a solid line for even–even nuclei, a dashed line for odd–odd nuclei and a dotted line for odd nuclei.

tively.) This pattern holds for all  $A$ , with breaks showing up in the parabolas when magic neutron or proton numbers are encountered.

Based on these observations, a pairing term is generally added to the mass formula in the form  $\alpha_p \delta(N, Z)/\sqrt{A}$ , where

$$\delta(N, Z) = [(-1)^N + (-1)^Z]/2 \quad (3.2)$$

takes the value  $+1$  for even–even nuclei,  $-1$  for odd–odd nuclei and  $0$  for odd nuclei. This form takes the odd- $A$  nuclei as the norm, with even–even nuclei more bound and odd–odd nuclei less bound, by the same amount.

### 3.5. Curvature term

If the semi-empirical mass formula is regarded as incorporating an expansion in powers of inverse radius,  $1/A^{1/3}$ , as exemplified by the volume and surface terms, then the next order would involve a term  $\alpha_R A^{1/3}$ , generally referred to as a curvature term. (The subscript  $R$  is chosen to recall the radius of curvature, since  $c$  might lead to confusion with the Coulomb term.) Inspection of the BW residuals in Fig. 5 reveals, in addition to the magic number cusps, a systematic decline at large  $A$ . The curvature term turns out to address this possible hint at omitted physics.

### 3.6. Shell effects

Reference has already been made to the parabolic form of the BW residuals between magic numbers, as exemplified by Fig. 5. A coarse-grained treatment of these shell effects can be based on Casten's observation [14] that widely scattered nuclear data tend to cluster significantly when plotted against the valence-nucleon variable

$$P = \frac{\nu_n \nu_p}{\nu_n + \nu_p}, \quad (3.3)$$

where  $\nu_n$  and  $\nu_p$  are the numbers of valence nucleons (the difference between the actual nucleon numbers,  $N$  and  $Z$  respectively, and the nearest magic numbers). The BW residuals are plotted

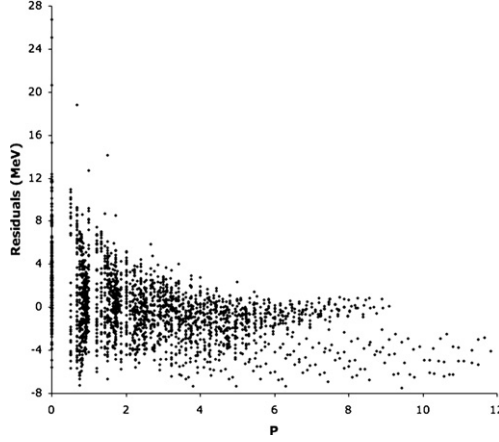


Fig. 8. Residuals of the BW fit to the mass table vs. the valence-nucleon variable  $P$  defined in Eq. (3.3).

against Casten's  $P$  in Fig. 8, which should be compared with Fig. 5. For the purpose of computing  $P$ , the magic numbers are taken to be the canonical 2, 8, 20, 28, 50, 82, 126, 184, for both neutrons and protons.

The residuals do become more compact in the  $P$  plot, and there is a suggestion of an overall parabolic behaviour. A correction term linear in  $P$  was shown some time ago [15] to improve the mass formula, but Fig. 8 suggests a different form. A global shell correction might be parametrised as  $\alpha_m P + \beta_m P^2$ . (Since the subscript  $p$  has been used for the pairing term, subscripts  $m$ , for “magic”, are preferred for the shell correction terms.) Note that, although it is referred to as a single term, the shell correction involves two fit coefficients.

### 3.7. Other terms

At various times, many other terms have been suggested for inclusion in the mass formula. The early example of the droplet model [8], which introduced a large number of possible candidates, has already been mentioned. To avoid excessive inflation of the discussion, the analysis here will be restricted to the terms described above. There are ten terms in all, the four BW terms and the six extra terms, involving a total of eleven fit coefficients. The full mass formula is

$$B = \alpha_v A + \alpha_s A^{2/3} + \alpha_c \frac{Z^2}{A^{1/3}} + \alpha_t \frac{(N-Z)^2}{A} + \alpha_{xc} \frac{Z^{4/3}}{A^{1/3}} + \alpha_w \frac{|N-Z|}{A} + \alpha_{st} \frac{(N-Z)^2}{A^{4/3}} + \alpha_p \frac{\delta(N, Z)}{\sqrt{A}} + \alpha_R A^{1/3} + \alpha_m P + \beta_m P^2, \quad (3.4)$$

where  $\delta(N, Z)$  is defined in Eq. (3.2) and  $P$  is defined in Eq. (3.3). The coefficients arising from a least-squares fit of the BW mass formula of Eq. (2.1) have been listed in Eq. (2.2). They may be compared with those obtained from a fit of the full mass formula of Eq. (3.4), displayed in Table 1. [The symmetry and Wigner terms in the table can be combined as a single term,  $\alpha_T T(T+a)/A$ , with  $\alpha_T = -126.05(108)$  MeV and  $a = 0.689(36)$ .] The improvement in  $D_{\text{rms}}$  is 1.916 MeV, a reduction of 54%.

It is of considerable interest that the fit coefficients obtained for the classical BW terms with the expanded formula differ substantially from those of the simple BW fit, shown in Eq. (2.2).

Table 1

Best fit values (in MeV) of coefficients of the semi-empirical mass formula of Eq. (3.4)

| $\alpha_v$    | $\alpha_s$ | $\alpha_C$ | $\alpha_t$ | $\alpha_{xC}$ | $\alpha_W$  |
|---------------|------------|------------|------------|---------------|-------------|
| 16.58(6)      | −26.95(35) | −0.774(2)  | −31.51(27) | 2.22(5)       | −43.40(222) |
| $\alpha_{st}$ | $\alpha_p$ | $\alpha_R$ | $\alpha_m$ | $\beta_m$     | rms         |
| 55.62(140)    | 9.87(41)   | 14.77(61)  | −1.90(5)   | 0.140(5)      | 1.620       |

The volume coefficient is more than an MeV larger, the surface coefficient 64% larger in magnitude, the Coulomb coefficient 12% larger in magnitude (but recall the balancing role of the exchange Coulomb term) and the symmetry coefficient 40% larger in magnitude than in the earlier fit. Expanding the mass formula to accommodate additional terms not only improves the fit to the measured masses but also plays an important role in determining the classical fit parameters.

#### 4. Mass formula fits

Starting from the basic BW mass formula, the extra terms (exchange Coulomb, Wigner, surface symmetry, pairing, curvature and shell correction) can be added individually or in various combinations, to test the effectiveness of each term and the way the different terms interact with one another. The basic tool for assessing the influence of different sets of extra terms is the change in rms deviation,  $\Delta_{\text{rms}}$ , as defined in Eq. (1.2), but a full understanding of the interplay of different terms requires a careful inspection of the numerical values of the coefficients determined by the fits. The six extra terms can be introduced singly (6 fits), in pairs (15 fits), in triples (20 fits), in fours (15 fits), in fives (6 fits) or all together (1 fit). Together with the BW fit, there are 64 fits all told.

##### 4.1. Root-mean-square deviations

The reduction in the rms deviation arising from the inclusion of a particular term is a measure of the importance of that term to the global mass systematics. This may be probed in a number of ways—by inspecting  $\Delta_{\text{rms}}$  upon adding only the term of interest to the BW mass formula, by noting the change in  $D_{\text{rms}}$  (defined in Eq. (1.1)) when the term alone is removed from the full mass formula (BW plus all six extra terms), or by analysing the effect of adding the term to all of the fits in which it does not appear. Any given extra term appears in 32 of the 64 fits performed. For every fit in which it appears, there will be a fit otherwise identical but not including it. The difference  $\delta(\text{rms})$  between the  $D_{\text{rms}}$  values of these two fits is a measure of the effectiveness of this term in improving the fit to the masses. (The differences are always taken so that  $\delta(\text{rms}) < 0$  if inclusion of the term under discussion reduces the overall rms deviation  $D_{\text{rms}}$ .) These are summarised in Table 2. For each extra term, the table lists  $\Delta_{\text{rms}}$  arising from the addition of that term to the BW formula; the minimum, maximum and mean changes in  $D_{\text{rms}}$  produced by adding the term to all 32 fits which do not contain it (and the standard deviation of the distribution); and the change in  $D_{\text{rms}}$  when the term is dropped from the full fit. (For brevity, generic changes in  $D_{\text{rms}}$  will be denoted  $\delta(\text{rms})$ .) The extra terms are identified by the subscripts on their fit coefficients, according to the key

$xC \Rightarrow$  exchange Coulomb,

$W \Rightarrow$  Wigner,

Table 2

Changes  $\delta_{\text{rms}}$  (in keV) in the rms deviations of fits associated with various extra terms in the mass formula. The first column identifies the extra term, using the key in Eq. (4.1). The second column shows  $\Delta_{\text{rms}}$  of Eq. (1.1). The next four columns list properties of the distribution of changes  $\delta(\text{rms})$  in  $D_{\text{rms}}$  arising from addition of the extra term to 32 fits not including it. The final column gives the increase in  $D_{\text{rms}}$  caused by removing the extra term from the full 10-term fit

| Extra term | $\Delta_{\text{rms}}$ | min    | max     | mean   | std. dev. | remove |
|------------|-----------------------|--------|---------|--------|-----------|--------|
| $xC$       | −88.1                 | −63.7  | −538.3  | −315.2 | 147.6     | 523.2  |
| $W$        | −283.5                | −4.7   | −651.1  | −265.8 | 218.4     | 133.9  |
| $st$       | −547.3                | −281.7 | −1017.8 | −536.8 | 231.5     | 500.7  |
| $p$        | −128.1                | −116.0 | −198.9  | −153.3 | 20.4      | 198.9  |
| $R$        | −26.8                 | 0.4    | −198.5  | −43.8  | 47.6      | 198.5  |
| $m$        | −507.2                | −498.2 | −769.0  | −584.1 | 78.8      | 769.0  |

$st \Rightarrow$  surface symmetry,

$p \Rightarrow$  pairing,

$R \Rightarrow$  curvature,

$m \Rightarrow$  shell correction.

(4.1)

The minimum, maximum, mean and standard deviation shown in each row of Table 2 relate to the distribution of the 32 values of  $\delta(\text{rms})$  from adding the corresponding term. All fits are bracketed by the two extremes—BW with no extra terms ( $D_{\text{rms}} = 3.536$  MeV) and the full fit, BW with all six extra terms ( $D_{\text{rms}} = 1.620$  MeV). Note that the inclusion of all six extra terms reduces the rms deviation by more than a factor of two (by almost 2 MeV).

By inspection of the  $\Delta_{\text{rms}}$  column in Table 2 it is clear that the most effective single term in improving the fit to the measured masses is the surface symmetry term (reducing the rms deviation by 547 keV), closely followed by the shell correction term (507 keV). Next in effectiveness is the Wigner term (284 keV). The pairing (128 keV) and exchange Coulomb (88 keV) terms make small improvements in the fit, while the curvature term (27 keV) has hardly any effect in reducing the rms deviation. Similar inspection of the “remove” column in Table 2 reveals that the greatest deterioration of the fit upon removing an individual term from the full mass formula occurs when the shell correction term is taken out (an increase in the rms deviation by 769 keV). Large changes are also occasioned by removal of the exchange Coulomb (523 keV) and surface symmetry (501 keV) terms, smaller changes by removal of the pairing (199 keV) and curvature (199 keV) terms, and the smallest change by removal of the Wigner (134 keV) term. These two columns alone highlight the important roles of the shell correction and surface symmetry terms and the relatively less important roles of the other terms in improving the overall fit, though the rather different effects of the exchange Coulomb term between the two extreme cases clearly suggests the probable significance of mutual interaction of different terms.

The four central columns of Table 2 embody the distribution of  $\delta(\text{rms})$  for each extra term over the 32 fits to which it can be added. The “min” and “max” columns together give the range of values of this quantity (where “max” refers to maximum effect, i.e. largest absolute value of  $\delta(\text{rms})$ , and “min” to minimum effect), while the mean and standard deviation characterize its size and spread. The extra terms may be considered in turn.

- The exchange Coulomb term has a fairly broad distribution of  $\delta(\text{rms})$  values, ranging from 64 to 538 keV, with a mean of 315 keV and a standard deviation 47% of the mean. The

implication is that the effect of this term is sensitive to the mix of terms included in the fit, but the overall size of its contribution to  $\Delta_{\text{rms}}$  supports its inclusion in the formula.

- The Wigner term, too, has a very broad distribution of  $\delta(\text{rms})$  values, ranging from 5 to 651 keV, with a mean of 266 keV and a standard deviation 82% of the mean. A glance at the 32 values of  $\delta(\text{rms})$  associated with the Wigner term immediately identifies significant clustering into two groups—16 values below 150 keV and another 16 above 280 keV. Even more striking, the former group of 16 small values includes all fits in which both the Wigner and surface symmetry terms are included, while the latter group of 16 large values includes all fits in which the Wigner term is included but the surface symmetry term is excluded (see Table 3). There is clearly a strong interplay between the two terms. The Wigner term plays an important role in improving the fit in the absence of a surface symmetry term, a marginal role in its presence.
- The surface symmetry term also displays a broad distribution of  $\delta(\text{rms})$  values, ranging from 282 to 1018 keV, with a mean of 537 keV and a standard deviation 43% of the mean, but is centered at large values. The strong interplay with the Wigner term, as outlined in Table 3, is again revealed in clustering into two groups—with and without a Wigner term—the former below 500 keV, the latter above 530 keV. The surface symmetry term is always important, but less so when a Wigner term is present.
- The pairing term is remarkably stable, with  $\delta(\text{rms})$  values ranging from 116 to 199 keV, with a mean of 153 keV and a standard deviation only 13% of the mean. It is almost impervious to the changing makeup of the mass formula, making a not very large but impressively robust contribution to the fit.
- The curvature term makes a rather small contribution to the fit, with  $\delta(\text{rms})$  values ranging from 0 to 199 keV, with a mean of 44 keV and a standard deviation 109% of the mean. Its inclusion in the mass formula may seem questionable, but it does have a not totally negligible effect when dropped from the full fit.
- Finally, the shell correction term has a narrow distribution of  $\delta(\text{rms})$  values, ranging from 498 to 769 keV, with a mean of 584 keV and a standard deviation a mere 13% of the mean. It is arguably the most important single term in the full mass formula, though it is occasionally eclipsed by the surface symmetry contribution, and is again remarkably robust.

Contemplation of the changes in  $D_{\text{rms}}$  arising from the inclusion of different mixtures of extra terms in the mass formula has revealed a certain hierarchy of importance (shell correction, surface symmetry, exchange Coulomb, Wigner, pairing and curvature), two very robust contributions (shell correction and pairing) and some striking correlations (Wigner and surface symmetry). The variability of  $\delta(\text{rms})$  with changes in the mix of terms suggests the presence of further

Table 3  
Comparison of  $\delta(\text{rms})$  distributions for fits including and/or excluding Wigner and surface symmetry terms. The table is similar to the four central columns of Table 2. The first column refers to inclusion (“in”) or exclusion (“out”) of the two terms considered. All entries in keV

| Extra term | min    | max     | mean   | std. dev. |
|------------|--------|---------|--------|-----------|
| W (st in)  | −4.7   | −149.1  | −77.0  | 49.5      |
| W (st out) | −283.5 | −651.1  | −454.6 | 141.9     |
| st (W in)  | −281.7 | −500.7  | −348.0 | 59.5      |
| st (W out) | −522.6 | −1017.8 | −725.6 | 176.6     |

correlations, which are revealed when attention is shifted to the values of the fit coefficients, rather than the global goodness of fit embodied in the rms deviation.

#### 4.2. Correlations

There are various possible ways of studying the presence of correlations between different terms of the semi-empirical mass formula. Formally, the preferred method is diagonalisation of the error matrix of the least-squares linear fit of the mass data to the formula (3.4). The mass formula is rephrased as

$$B = \sum_X \alpha_X F_X, \quad (4.2)$$

which serves, by comparison with Eq. (3.4), to define the functional forms  $F_X$ , each a function of  $Z, N, A$ . The sum runs over  $X = v, s, C, t, xC, W, st, p, R, m, m_2$ , with  $\alpha_{m_2}$  representing  $\beta_m$ . The matrix  $\mathcal{R}$  is defined by its matrix elements,

$$\mathcal{R}_{XY} = \sum_i F_X(i) F_Y(i), \quad (4.3)$$

the sum running over all 2225 data points  $i$ , and the error matrix is equal to the inverse,  $\mathcal{R}^{-1}$ . The standard error in each fit coefficient is determined from the corresponding diagonal term of the error matrix, being equal to the product of the square root of the diagonal matrix element with the rms deviation of the fit. The off-diagonal elements of the error matrix encode the correlations between the coefficients.

Diagonalising the error matrix produces, as eigenvectors, linear combinations of fit coefficients which are uncorrelated with one another. The extent to which the values of these linear combinations are constrained is fixed by the eigenvalues associated with each eigenvector—the square root of the eigenvalue is a measure of the standard error in the eigenvector. The independent linear combinations of fit coefficients so found are listed in Table 4, together with the square roots of their error matrix eigenvalues. As always, the eigenvectors are determined up to an overall normalisation factor. The combinations shown in the table are normalised so the sums of the squares of the amplitudes of different fit coefficients in each combination is unity. Only amplitudes exceeding 0.1 are shown and terms are listed in decreasing order of amplitude.

Table 4

Independent linear combinations of fit parameters found by diagonalising the error matrix. The column labeled “constraint” gives the square root of the corresponding eigenvalue of the error matrix

| Eigenvector                                                                                                    | Constraint |
|----------------------------------------------------------------------------------------------------------------|------------|
| $0.90\alpha_v + 0.28\alpha_{xC} + 0.27\alpha_s - 0.19\alpha_C$                                                 | 0.000862   |
| $0.71\alpha_s + 0.49\alpha_R + 0.34\alpha_t - 0.23\alpha_m - 0.20\alpha_v - 0.19\alpha_{xC} + 0.11\alpha_{st}$ | 0.0245     |
| $0.98\alpha_C + 0.18\alpha_v$                                                                                  | 0.000026   |
| $0.77\alpha_t + 0.57\alpha_{xC} - 0.22\alpha_s - 0.14\alpha_v + 0.10\alpha_{st}$                               | 0.0512     |
| $0.73\alpha_{xC} - 0.50\alpha_t - 0.31\alpha_v + 0.30\alpha_s + 0.13\alpha_R$                                  | 0.00531    |
| $0.84\alpha_W - 0.51\alpha_{st} - 0.17\alpha_R$                                                                | 1.636      |
| $0.78\alpha_{st} + 0.39\alpha_W - 0.39\alpha_R + 0.23\alpha_p + 0.17\alpha_s$                                  | 0.241      |
| $0.97\alpha_p - 0.21\alpha_{st} - 0.12\alpha_W$                                                                | 0.254      |
| $0.73\alpha_R - 0.48\alpha_s + 0.37\alpha_W + 0.25\alpha_{st} - 0.13\alpha_t$                                  | 0.324      |
| $0.96\alpha_m + 0.14\alpha_s - 0.13\alpha_{xC} + 0.11\alpha_R - 0.10\beta_m$                                   | 0.0280     |
| $0.99\beta_m + 0.097\alpha_m$                                                                                  | 0.00114    |

Linear combinations in which the leading term has an amplitude exceeding 0.9 are those dominated by the volume, Coulomb, pairing and shell correction terms, respectively. They involve only small admixtures of other terms and are tightly constrained in their ranges of values, except for the pairing-dominated term, where the constraint is much looser. All other combinations involve significant mixtures of different terms. There are two quite tightly constrained combinations of symmetry and exchange Coulomb terms, with smaller admixtures of surface and volume terms. One combination of surface and curvature terms (with admixtures of several other terms) is quite well constrained, while the other combination of these two terms (with admixtures of yet other terms) is relatively lightly constrained. Finally, two loosely constrained combinations of surface symmetry and Wigner terms are somewhat influenced by the curvature term.

On the basis of this analysis, it is to be expected that there will not be much correlation of the volume, Coulomb, pairing and shell correction terms with other terms, while significant correlation is anticipated between symmetry and exchange Coulomb terms, between surface and curvature terms and between Wigner and surface symmetry terms.

An alternative signature of correlation can be obtained by plotting against one another the values of the fit coefficients of a pair of mass formula terms in all fits containing both terms. A linear trendline is fitted to each such scatter plot and a trend coefficient  $R^2$  computed (the scaled rms deviation of the points in the scatter plot from the trendline). Correlation plots may be drawn for all pairs of fit coefficients, 55 plots in all. They fall into classes, which may be characterised as showing strong correlation ( $0.9 < R^2 < 1$ ), moderate correlation ( $0.6 < R^2 < 0.9$ ), weak correlation ( $0.2 < R^2 < 0.6$ ) or no correlation ( $0 < R^2 < 0.2$ ). However, these characterisations should be used with caution, since the scatter plots are more varied than can be described by a single linear trend parameter. For instance, a plot ostensibly showing no correlation might have three narrow horizontal bands of points, suggesting a more subtle hidden correlation. Representative examples of these classes of correlation plot, all involving the coefficient of the symmetry term,  $\alpha_I$ , are illustrated in Fig. 9.

The correlation plot for a pair of BW terms contains 64 points (since both terms occur in all 64 fits); the correlation plot between a BW term and an extra term contains 32 points (since each extra term occurs in 32 fits, every BW term in all 64 fits); while the correlation plot between two extra terms contains only 16 points (since each extra term occurs in 32 fits, half of which contain the second extra term and half not). The trend coefficients for all 55 correlation plots are listed in Table 5. Of the 55 correlation plots, there are 5 showing strong correlation and 13 showing no correlation. According to this mechanical criterion, there are strong correlations between the curvature and volume coefficients; between the curvature and surface coefficients; between the Wigner and symmetry coefficients; between the Wigner and surface symmetry coefficients; and between the linear and quadratic shell correction coefficients. There is no correlation between the pairing and any other coefficient, nor between the Wigner and Coulomb coefficients; between the Wigner and exchange Coulomb coefficients; or between the surface symmetry and exchange Coulomb coefficients. There is weak to moderate correlation between all other pairs of coefficients.

Further correlations are revealed by a careful study of the fit coefficients and their dependence on the selection of terms to be included in the fit.

#### 4.3. Fit coefficients

The changes in the four standard BW fit parameters when the extra terms are added individually, producing five-parameter mass formulas (six parameters in the case of the shell correction



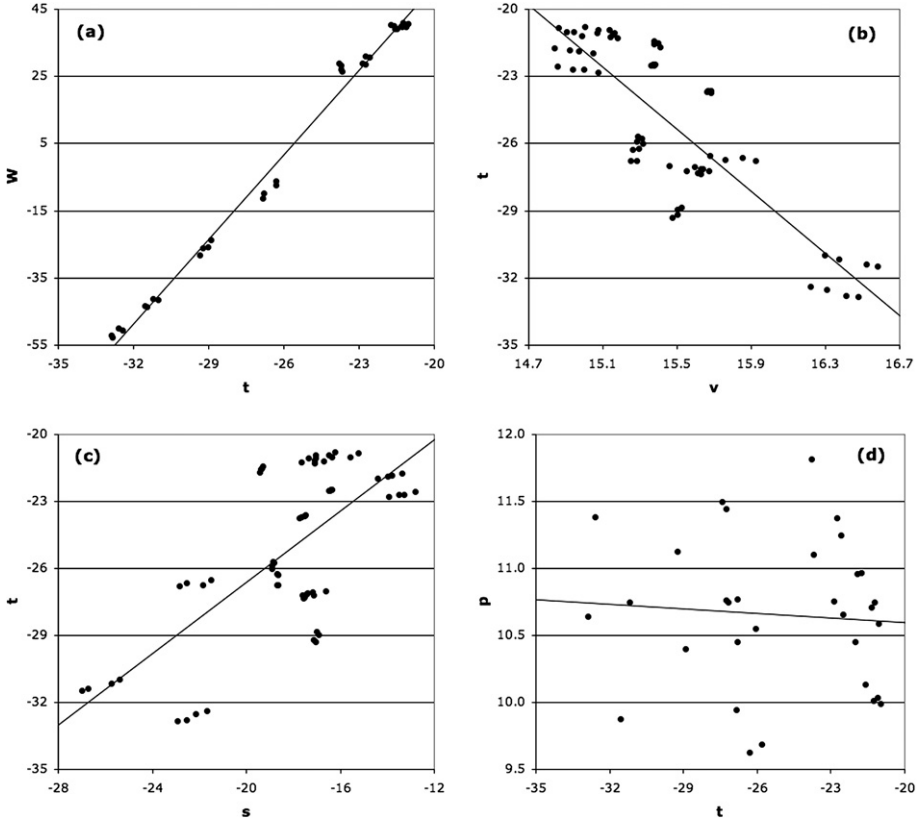


Fig. 9. Representative correlation plots for (a) strong [ $R^2 = 0.99$ ], (b) moderate [ $R^2 = 0.72$ ], (c) weak [ $R^2 = 0.48$ ] and (d) no [ $R^2 = 0.01$ ] correlation. All coefficients in MeV. The line in each plot is a linear trendline.

Table 5

Trend coefficients  $R^2$  for all pairwise correlation plots between coefficients of the full mass formula, Eq. (3.4). Terms are identified by the subscripts on their coefficients. The table is symmetric about the (blank) main diagonal

|       | $v$  | $s$  | $C$  | $t$  | $xC$ | $W$  | $st$ | $p$  | $R$  | $m$  | $m_2$ |
|-------|------|------|------|------|------|------|------|------|------|------|-------|
| $v$   |      | 0.76 | 0.42 | 0.72 | 0.27 | 0.62 | 0.63 | 0.08 | 0.93 | 0.44 | 0.48  |
| $s$   | 0.76 |      | 0.85 | 0.48 | 0.36 | 0.46 | 0.58 | 0.06 | 0.95 | 0.69 | 0.60  |
| $C$   | 0.42 | 0.85 |      | 0.18 | 0.63 | 0.10 | 0.27 | 0.19 | 0.73 | 0.71 | 0.57  |
| $t$   | 0.72 | 0.48 | 0.18 |      | 0.27 | 0.99 | 0.82 | 0.01 | 0.79 | 0.46 | 0.59  |
| $xC$  | 0.27 | 0.36 | 0.63 | 0.27 |      | 0.04 | 0.11 | 0.02 | 0.23 | 0.78 | 0.79  |
| $W$   | 0.62 | 0.46 | 0.10 | 0.99 | 0.04 |      | 0.90 | 0.01 | 0.89 | 0.38 | 0.61  |
| $st$  | 0.63 | 0.58 | 0.27 | 0.82 | 0.11 | 0.90 |      | 0.00 | 0.86 | 0.34 | 0.46  |
| $p$   | 0.08 | 0.06 | 0.19 | 0.01 | 0.02 | 0.01 | 0.00 |      | 0.06 | 0.20 | 0.10  |
| $R$   | 0.93 | 0.95 | 0.73 | 0.79 | 0.23 | 0.89 | 0.86 | 0.06 |      | 0.67 | 0.65  |
| $m$   | 0.44 | 0.69 | 0.71 | 0.46 | 0.78 | 0.38 | 0.34 | 0.20 | 0.67 |      | 0.96  |
| $m_2$ | 0.48 | 0.60 | 0.57 | 0.59 | 0.79 | 0.61 | 0.46 | 0.10 | 0.65 | 0.96 |       |

term), are evident by inspection of Table 6. The coefficients are not strongly affected by the inclusion of the additional term, with the exception of the curvature term, which, despite its overall small effect on the rms deviation, strongly influences the surface coefficient and shifts the

Table 6

Least-squares fits of five-term mass formulae to binding energies. Columns list the coefficients of the mass formula (in MeV) and the rms deviation of the fit (in keV). The coefficient of the extra term is denoted  $\alpha_X$  and the extra term is identified by the key in Eq. (4.1). (For the shell correction there are two coefficients,  $\alpha_m$  above,  $\beta_m$  below)

| $X$  | $\alpha_v$ | $\alpha_s$ | $\alpha_C$ | $\alpha_t$ | $\alpha_X$ | rms  |
|------|------------|------------|------------|------------|------------|------|
| $BW$ | 15.36(3)   | −16.42(8)  | −0.691(2)  | −22.53(7)  |            | 3536 |
| $xC$ | 15.15(3)   | −17.06(10) | −0.702(2)  | −21.19(14) | 1.12(10)   | 3448 |
| $W$  | 15.67(3)   | −17.62(10) | −0.705(2)  | −23.71(9)  | 27.98(139) | 3253 |
| $st$ | 15.62(2)   | −17.50(8)  | −0.698(2)  | −27.33(17) | 23.92(80)  | 2989 |
| $p$  | 15.38(3)   | −16.47(8)  | −0.692(2)  | −22.55(7)  | 11.24(86)  | 3408 |
| $R$  | 14.84(9)   | −13.34(53) | −0.669(4)  | −21.75(15) | −5.26(89)  | 3510 |
| $m$  | 15.38(2)   | −16.30(7)  | −0.693(2)  | −22.48(6)  | −1.52(8)   | 3029 |
|      |            |            |            |            | 0.077(9)   |      |

Table 7

Values (in MeV) of BW coefficients and rms deviations when individual extra terms are omitted from the semi-empirical mass formula of Eq. (3.4). See Eq. (4.1) for key to extra terms

| Omit | $\alpha_v$ | $\alpha_s$ | $\alpha_C$ | $\alpha_t$ | rms   |
|------|------------|------------|------------|------------|-------|
|      | 16.58(6)   | −26.95(35) | −0.774(2)  | −31.51(27) | 1.620 |
| $xC$ | 16.48(7)   | −22.89(45) | −0.729(3)  | −32.87(36) | 2.144 |
| $W$  | 15.93(5)   | −22.79(30) | −0.755(2)  | −26.81(14) | 1.754 |
| $st$ | 15.14(6)   | −17.62(34) | −0.727(3)  | −21.25(11) | 2.121 |
| $p$  | 16.52(6)   | −26.72(40) | −0.773(3)  | −31.42(30) | 1.819 |
| $R$  | 15.30(2)   | −18.65(8)  | −0.723(1)  | −26.27(18) | 1.819 |
| $m$  | 16.38(8)   | −25.72(51) | −0.760(4)  | −31.17(39) | 2.389 |

volume and Coulomb coefficients a fair bit, and of the surface symmetry term, which significantly increases the magnitude of the standard symmetry coefficient. Note that  $\alpha_W$  enters with the opposite sign to that expected, corresponding to a  $T(T - 0.59)$  behaviour when added to the symmetry term.

It is also instructive to show the effect on the full fit to Eq. (3.4) of omitting a single term at a time. This is shown in Table 7. Leaving out only the exchange Coulomb term costs 524 keV in rms deviation and reduces noticeably the magnitude of  $\alpha_s$ . Omitting only the Wigner term costs only 134 keV in rms deviation while reducing the magnitudes of  $\alpha_v$ ,  $\alpha_s$  and  $\alpha_t$ . Discarding only the surface symmetry term costs 501 keV in rms deviation and strongly reduces the magnitudes of  $\alpha_v$ ,  $\alpha_s$  and  $\alpha_t$ . Dispensing with only the pairing term costs just 199 keV in rms deviation while leaving the BW coefficients almost unchanged. Removing only the curvature term costs only 199 keV in rms deviation while reducing markedly the magnitudes of all BW coefficients. Finally, dropping only the shell correction terms costs 769 keV in rms deviation but leaves the BW coefficients practically unchanged.

The implication is that the pairing and shell correction terms have almost no effect on determining the values of the BW coefficients, but contribute to the reduction of the rms deviation, the shell correction being particularly important in this respect. The exchange Coulomb and surface symmetry terms are also significant in reducing the rms deviation, while the surface symmetry and curvature terms play important roles in assigning the final values to the BW coefficients. From this it might be concluded that the pairing and shell correction terms are not correlated with the other extra terms, but that these other terms are substantially correlated with one another.

Table 8

Characteristics of the distribution of values of mass formula coefficients for each term in the formula over the set of fits in which the term participates (64 fits for each BW term, 32 fits for each extra term). The mean, standard deviation, maximum and minimum (all in MeV) are shown for each term

| Coefft.       | mean   | std. dev. | max    | min    |
|---------------|--------|-----------|--------|--------|
| $\alpha_v$    | 15.46  | 0.45      | 16.58  | 14.84  |
| $\alpha_s$    | −18.13 | 3.16      | −12.78 | −26.95 |
| $\alpha_C$    | −0.711 | 0.025     | −0.669 | −0.774 |
| $\alpha_t$    | −25.13 | 3.67      | −20.82 | −32.87 |
| $\alpha_{xC}$ | 1.88   | 0.46      | 2.38   | 0.97   |
| $\alpha_W$    | 0.95   | 35.99     | 40.75  | −53.01 |
| $\alpha_{st}$ | 35.73  | 12.02     | 55.91  | 22.70  |
| $\alpha_p$    | 10.66  | 0.55      | 11.81  | 9.62   |
| $\alpha_R$    | 1.28   | 7.07      | 14.77  | −8.42  |
| $\alpha_m$    | −1.67  | 0.14      | −1.42  | −1.96  |
| $\beta_m$     | 0.106  | 0.023     | 0.147  | 0.067  |

The distribution of values of the various fit coefficients over the different fits in which each term of the mass formula participates is summarised in Table 8. It is evident that some terms are relatively independent of the choice of other terms to join them in the mass formula, while some terms are quite sensitive to this choice and others vary drastically as the selection of terms included changes. The robust terms are the volume, Coulomb, pairing and linear shell correction terms, for all of which the standard deviation of their values is less than 10% of their mean value. The more sensitive terms are the surface, symmetry, exchange Coulomb, surface symmetry and quadratic shell correction terms, for all of which the standard deviation of their values lies between 14% and 35% of their mean value. The drastically variable terms are the Wigner and curvature terms, for which even their sign is not definite. Significant correlations between some terms are clearly suggested.

A detailed picture of correlations between terms is afforded by listing a particular coefficient in ascending order, together with the identification of the extra terms included in the fit for each value. This is illustrated in Table 9 for the specific case of the Wigner coefficient  $\alpha_W$ . It is evident from this table, too, that this coefficient is very sensitive to the precise make-up of the mass formula, ranging in value from −53 MeV to 41 MeV. There is also noticeable clustering of the other extra terms in the table. The most prominent result is that  $\alpha_W$  is negative in all fits which include the surface symmetry term, positive in all fits which do not include the surface symmetry term. Since the symmetry coefficient  $\alpha_t$  is negative in all fits, a  $T(T + a)$  form of the mass formula would involve  $a < 0$  for  $\alpha_W > 0$ , a most peculiar and unwelcome feature. It is tempting to define positive  $\alpha_W$  as “wrong” and to rule out any mass formula which does not contain a surface symmetry term.

In the other columns of Table 9, the check marks are fairly randomly scattered in the pairing and shell correction columns, indicating that the Wigner term is not correlated with these two terms, but they cluster in the exchange Coulomb column and, when  $\alpha_W < 0$ , in the curvature column. Concentrating on the “correct” sign of  $\alpha_W$ , the Wigner coefficient is large in absolute value when a curvature term is present, relatively small when the curvature term is absent. In the absence of a curvature term,  $|\alpha_W|$  is very small when the exchange Coulomb term is included.

A similar picture is obtained if, instead of the value of  $\alpha_W$ , the reduction in rms deviation  $\Delta_{\text{rms}}$  due to the Wigner term is listed in ascending order of importance, as shown in Table 10. Again, as previously noted, there is a dramatic correlation between the Wigner and surface symmetry

Table 9  
Values (in MeV) of the Wigner coefficient  $\alpha_W$  from all 32 fits including a Wigner term, in ascending order. The other extra terms participating in each fit are indicated by check marks  $\checkmark$  in the corresponding columns

| $\alpha_W$ | $xC$         | $st$         | $p$          | $R$          | $m$          | $\alpha_W$ | $xC$         | $st$ | $p$          | $R$          | $m$          |
|------------|--------------|--------------|--------------|--------------|--------------|------------|--------------|------|--------------|--------------|--------------|
| −53.01     |              | $\checkmark$ |              | $\checkmark$ | $\checkmark$ | 26.37      |              |      |              |              | $\checkmark$ |
| −52.27     |              | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | 26.91      |              |      | $\checkmark$ |              | $\checkmark$ |
| −50.70     |              | $\checkmark$ |              | $\checkmark$ |              | 27.98      |              |      |              |              |              |
| −50.29     |              | $\checkmark$ | $\checkmark$ | $\checkmark$ |              | 28.43      |              |      |              | $\checkmark$ | $\checkmark$ |
| −43.85     | $\checkmark$ | $\checkmark$ |              | $\checkmark$ | $\checkmark$ | 28.66      |              |      | $\checkmark$ |              |              |
| −43.40     | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | 28.77      |              |      | $\checkmark$ | $\checkmark$ | $\checkmark$ |
| −41.58     | $\checkmark$ | $\checkmark$ |              | $\checkmark$ |              | 30.47      |              |      |              | $\checkmark$ |              |
| −41.44     | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ |              | 30.93      |              |      | $\checkmark$ | $\checkmark$ |              |
| −28.47     |              | $\checkmark$ |              |              |              | 38.99      | $\checkmark$ |      |              |              | $\checkmark$ |
| −26.30     |              | $\checkmark$ | $\checkmark$ |              |              | 39.08      | $\checkmark$ |      | $\checkmark$ |              | $\checkmark$ |
| −26.05     |              | $\checkmark$ |              |              | $\checkmark$ | 39.49      | $\checkmark$ |      |              | $\checkmark$ | $\checkmark$ |
| −23.99     |              | $\checkmark$ | $\checkmark$ |              | $\checkmark$ | 39.51      | $\checkmark$ |      | $\checkmark$ | $\checkmark$ | $\checkmark$ |
| −11.35     | $\checkmark$ | $\checkmark$ |              |              |              | 39.81      | $\checkmark$ |      |              |              |              |
| −9.88      | $\checkmark$ | $\checkmark$ | $\checkmark$ |              |              | 40.04      | $\checkmark$ |      | $\checkmark$ |              |              |
| −7.67      | $\checkmark$ | $\checkmark$ |              |              | $\checkmark$ | 40.61      | $\checkmark$ |      |              | $\checkmark$ |              |
| −6.32      | $\checkmark$ | $\checkmark$ | $\checkmark$ |              | $\checkmark$ | 40.75      | $\checkmark$ |      | $\checkmark$ | $\checkmark$ |              |

Table 10  
Values (in keV) of the reduction  $\Delta_{\text{rms}}$  of rms deviation upon adding a Wigner term to 32 fits without a Wigner term, in ascending order. The extra terms to which the Wigner term is added in each fit are indicated by check marks  $\checkmark$  in the corresponding columns

| $\Delta_{\text{rms}}$ | $xC$         | $st$         | $p$          | $R$          | $m$          | $\Delta_{\text{rms}}$ | $xC$         | $st$ | $p$          | $R$          | $m$          |
|-----------------------|--------------|--------------|--------------|--------------|--------------|-----------------------|--------------|------|--------------|--------------|--------------|
| −4.7                  | $\checkmark$ | $\checkmark$ | $\checkmark$ |              | $\checkmark$ | −283.5                |              |      |              |              |              |
| −6.4                  | $\checkmark$ | $\checkmark$ |              |              | $\checkmark$ | −289.5                |              |      |              |              | $\checkmark$ |
| −8.6                  | $\checkmark$ | $\checkmark$ | $\checkmark$ |              |              | −310.2                |              |      | $\checkmark$ |              |              |
| −10.8                 | $\checkmark$ | $\checkmark$ |              |              |              | −317.9                |              |      | $\checkmark$ |              | $\checkmark$ |
| −62.6                 |              | $\checkmark$ | $\checkmark$ |              |              | −330.9                |              |      |              | $\checkmark$ | $\checkmark$ |
| −64.5                 |              | $\checkmark$ | $\checkmark$ |              | $\checkmark$ | −331.0                |              |      |              | $\checkmark$ |              |
| −69.7                 |              | $\checkmark$ |              |              |              | −355.2                |              |      | $\checkmark$ | $\checkmark$ |              |
| −71.0                 |              | $\checkmark$ |              |              | $\checkmark$ | −356.9                |              |      | $\checkmark$ | $\checkmark$ | $\checkmark$ |
| −80.9                 | $\checkmark$ | $\checkmark$ |              | $\checkmark$ |              | −523.4                | $\checkmark$ |      |              |              |              |
| −85.8                 | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ |              | −543.8                | $\checkmark$ |      |              | $\checkmark$ |              |
| −107.7                |              | $\checkmark$ |              | $\checkmark$ |              | −552.9                | $\checkmark$ |      | $\checkmark$ |              |              |
| −112.1                |              | $\checkmark$ | $\checkmark$ | $\checkmark$ |              | −571.2                | $\checkmark$ |      | $\checkmark$ | $\checkmark$ |              |
| −122.5                | $\checkmark$ | $\checkmark$ |              | $\checkmark$ | $\checkmark$ | −601.4                | $\checkmark$ |      |              |              | $\checkmark$ |
| −133.9                | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | −614.6                | $\checkmark$ |      |              | $\checkmark$ | $\checkmark$ |
| −142.1                |              | $\checkmark$ |              | $\checkmark$ | $\checkmark$ | −639.9                | $\checkmark$ |      | $\checkmark$ |              | $\checkmark$ |
| −149.1                |              | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | −651.1                | $\checkmark$ |      | $\checkmark$ | $\checkmark$ | $\checkmark$ |

terms— $\Delta_{\text{rms}}$  from adding a Wigner term is between 5 and 150 keV whenever a surface symmetry term is included in the fit, between 280 and 650 keV whenever no surface symmetry term is included. The random distribution of check marks in the pairing and shell correction columns suggests no correlation of these two terms with the Wigner term. The curvature term tends to increase the effectiveness of the Wigner term when there is a surface symmetry term present, but has no clear effect in its absence. Without a surface symmetry term, the presence of an exchange Coulomb term is associated with the largest Wigner effects, with  $\Delta_{\text{rms}}$  exceeding 520 keV when and only when exchange Coulomb is included.

This analysis reveals important correlations between the Wigner term and the surface symmetry, curvature and exchange Coulomb terms. The first of these was found also in the error matrix and correlation plot analyses, but the second did not show up in either of these analyses and the third was directly contradicted by the correlation plot analysis. The interplay between terms of the mass formula appears to be too subtle to be treated by a single correlation criterion.

Analyses of the fit coefficients and  $\Delta_{\text{rms}}$  values similar to those carried out here for the Wigner term can be applied to all the terms of the mass formula (though only the former is relevant to the four BW terms). The results are as follows, with the robust terms treated first, then the variable terms and finally the highly sensitive terms.

- The volume term is robust, as noted, with  $\alpha_v$  taking values between 14.84 and 16.58 MeV. It exceeds 16 MeV when and only when Wigner, surface symmetry and curvature terms are all present.
- Also robust is the Coulomb term, with  $\alpha_C$  ranging from  $-0.669$  to  $-0.774$  MeV. When, and only when, exchange Coulomb, surface symmetry and curvature terms are included in the fit,  $|\alpha_C| > 0.740$  MeV.
- The robust pairing term coefficient  $\alpha_p$  is between 9.62 and 11.81 MeV. It exceeds 11 MeV only when the exchange Coulomb term is omitted. Its contribution of 116 to 199 keV to reduction of the rms deviation of the fit tends to be larger when surface symmetry or Wigner terms are included.
- The coefficient of the fourth robust term, the linear shell correction term, lies between  $-1.42$  and  $-1.96$  MeV and  $|\alpha_m| > 1.64$  MeV whenever the surface symmetry term is present.
- The surface coefficient  $\alpha_s$  varies over a rather wide range, from  $-12.78$  to  $-26.95$  MeV. In the BW fit, it takes the value  $-16.42$  MeV. Its absolute value exceeds 20 MeV only (and always) when surface symmetry, curvature and Wigner or exchange Coulomb terms are present. In all 17 fits with  $|\alpha_s| < 16.5$  MeV, the surface symmetry term is absent, and in 14 of these the curvature term is present.
- The symmetry term has a coefficient varying between  $-20.82$  and  $-32.87$  MeV. Its absolute value is less than 24 MeV in all fits which do not include a surface symmetry term, greater than 25.7 MeV in all fits which do include such a term, and is larger than 30 MeV when and only when surface symmetry, Wigner and curvature terms are all included.
- The coefficient of the exchange Coulomb term lies between 0.97 and 2.38 MeV. It is greater than 1.86 MeV in all fits with either a Wigner or a surface symmetry term, or both, and less than 1.27 MeV in all fits with neither of these two terms. (According to Table 5, it is moderately correlated with the Coulomb coefficient.) The contribution of this term to reducing the rms deviation of the fit is more than 400 keV when the fit includes the shell correction term, unless both the Wigner and surface symmetry terms are omitted, when the contribution drops to about 120 keV. It is between 270 and 540 keV as long as a Wigner or surface symmetry term is present.
- The surface symmetry term is also quite variable, its coefficient ranging from 22.70 to 55.91 MeV. All 16 fits without a Wigner term have  $\alpha_{st} < 31$  MeV; all 16 fits with a Wigner term have  $\alpha_{st} > 31$  MeV. When, and only when, both Wigner and curvature terms are present,  $\alpha_{st} > 53$  MeV. The reduction in rms deviation from inclusion of this term is between 280 and 500 keV if a Wigner term is included, between 520 and 1020 keV if the Wigner term is excluded. Without a Wigner term, but with an exchange Coulomb term, this contribution is 800 to 1020 keV.

- The quadratic shell correction term has a small coefficient, between 0.067 and 0.147 MeV, which is larger than 0.107 MeV whenever a surface symmetry term is included in the fit.
- The curvature coefficient is poorly determined, even its sign being variable, with  $\alpha_R < 0$  in 18 fits and  $\alpha_R > 0$  in 14 fits. The coefficient ranges from  $-8.42$  to  $14.77$  MeV. It is positive in all but three fits with a surface symmetry term and negative in all fits without a surface symmetry term. Its value is greater than 8 MeV when and only when both surface symmetry and Wigner terms are present, and exceeds 12 MeV if an exchange Coulomb term is added to these two. The curvature term reduces the rms deviation by less than 100 keV in all but three fits, the reduction being 90 to 200 keV in all four fits containing exchange Coulomb, Wigner and surface symmetry terms.
- The Wigner term has been discussed at length above.

## 5. Refinements

Various suggestions have been made for alternative treatment of some of the terms introduced here. The effect of some such alternatives has been tested in the context of the full fit described by Eq. (3.4) and the results are briefly described in this section.

### 5.1. Exponential Wigner term

It was pointed out in Section 3.2 that an exponential form of the Wigner term was advocated by Myers and Swiatecki [12], with  $\alpha_W|N - Z|/A$  being replaced by  $\alpha'_W \exp(-\beta_W|N - Z|/A)$ . A fit with this replacement was carried out, treating  $\alpha'_W$  as a linear fit parameter. The non-linear parameter  $\beta_W$  was scanned over the range  $1 \leq \beta_W \leq 4$ , a linear fit to all other parameters being performed for each  $\beta_W$  value. The rms deviation of the linear fit was found to be an extremely shallow function of  $\beta_W$ , with a minimum of 1.576 MeV at  $\beta_W = 2.55$ . This is smaller by only some 45 keV than the rms deviation of the standard fit, not a significant improvement.

Although the variation in minimum rms deviation with  $\beta_W$ , in the range  $2 \leq \beta_W \leq 4$ , was only 2–3 keV, some of the fit coefficients varied quite dramatically. This is illustrated in Table 11. The coefficient of the surface term ranges from  $-18.0$  MeV to  $-20.8$  MeV, the coefficient of the curvature term from  $-10.9$  MeV to  $-1.4$  MeV and the coefficient of the exponential Wigner term itself from 26.0 to 15.1 MeV. In addition, several of the fit coefficients differ quite strongly from their values in the linear Wigner fit, notably the surface, symmetry, exchange Coulomb, surface symmetry and (especially) curvature coefficients. The best fit coefficients depend not only on the selection of extra terms to be added to the BW mass formula but also on the specific form of the extra terms.

Given the marginal improvement in rms deviation, there does not seem to be compelling reason to forego the linear form of the Wigner term, with the concomitant loss of the option of the  $T(T + a)$  parametrisation, but the lesson of this exercise is that some of the fit coefficients are, in fact, much less well determined than is suggested by their standard errors.

### 5.2. Nonlinear surface symmetry term

In Section 3.3, reference was made to the non-linear surface symmetry correction advocated on thermodynamic grounds in Ref. [13]. The form there argued for is

$$\frac{(N - Z)^2}{A(1 + bA^{-1/3})},$$

Table 11

Fit coefficients when an exponential Wigner term is used. (See text for the form and parametrisation of this term.) Values are listed for different choices of the dimensionless non-linear fit parameter  $\beta_W$  and for the “standard” best fit of Table 1. All entries in MeV. The linear Wigner coefficient from the standard fit is in brackets

| Coefficient   | $\beta_W = 2$ | $\beta_W = 2.55$ | $\beta_W = 4$ | Standard      |
|---------------|---------------|------------------|---------------|---------------|
| $\alpha_v$    | 15.67(5)      | 15.81(4)         | 15.98(4)      | 16.58(6)      |
| $\alpha_s$    | −17.98(34)    | −19.20(31)       | −20.82(29)    | −26.95(35)    |
| $\alpha_C$    | −0.741(2)     | −0.745(2)        | −0.750(2)     | −0.774(2)     |
| $\alpha_t$    | −29.35(16)    | −29.50(17)       | −29.49(17)    | −31.51(27)    |
| $\alpha_{xC}$ | 1.57(6)       | 1.58(6)          | 1.59(6)       | 2.22(5)       |
| $\alpha'_W$   | 26.02(113)    | 21.64(94)        | 15.14(66)     | [−43.40(220)] |
| $\alpha_{st}$ | 44.71(79)     | 44.73(79)        | 43.24(75)     | 55.62(140)    |
| $\alpha_p$    | 10.38(40)     | 10.37(40)        | 10.36(40)     | 9.87(41)      |
| $\alpha_R$    | −10.89(87)    | −6.94(72)        | −1.43(55)     | 14.77(61)     |
| $\alpha_m$    | −1.86(4)      | −1.86(4)         | −1.85(4)      | −1.90(5)      |
| $\beta_m$     | 0.135(5)      | 0.134(5)         | 0.133(5)      | 0.140(5)      |
| rms           | 1.577         | 1.576            | 1.579         | 1.620         |

which encompasses both the regular symmetry term (when  $b = 0$ ) and a non-linear surface correction controlled by the non-linear parameter  $b$ . Once again, a fit including such a term was performed by scanning a range of values of  $b$  and carrying out a linear fit for each value, then seeking the minimum rms deviation as a function of  $b$ . The rather shallow minimum, 1.743 MeV, occurs at  $b = 3.6$  and is higher by 123 keV than the best fit result with a linear surface symmetry term. The rms deviation varies by 15 keV or less as  $b$  ranges from 3 to 5, but once again some of the fit coefficients change quite seriously over this range of  $b$ . (In particular, the coefficient of the symmetry term ranges from −33 MeV to −40 MeV, the coefficient of the Wigner term from −6 MeV to −17 MeV.) The fit alone supplies no compelling reason to prefer the non-linear form of the surface symmetry term to the linear form.

There is not much sensitivity of the fit coefficients to the choice of surface symmetry term. The volume, surface, Coulomb, exchange Coulomb, pairing and shell correction coefficients have very similar values for both the linear and non-linear fits, while the curvature (11.86 MeV non-linear, 14.77 MeV linear) and symmetry (−34.91 MeV non-linear, −31.51 MeV linear) coefficients do change somewhat and the Wigner coefficient changes drastically, from −43.40 MeV in the linear fit to −10.07 MeV in the non-linear fit. Once again the unavoidable conclusion is that the uncertainty in some of the coefficients considerably exceeds the standard error.

### 5.3. Uniform $T(T + a)$ dependence

With the symmetry and Wigner terms unified in a  $T(T + a)$  dependence, it would seem reasonable to apply a similar modification to the surface symmetry term—the  $\alpha_t \cdot 4T(T + a)/A$  term would be accompanied by an  $\alpha_{st} \cdot 4T(T + a)/A^{4/3}$  term. This is equivalent to introducing a surface Wigner term, though constrained by the restriction to a common value of  $a$ . The best fit is obtained with  $a = 1.2$ , producing a minimum rms deviation of 1.620 MeV, an improvement of only 0.3 keV. This certainly does not justify the inclusion of the additional term. However, in the presence of this term some of the fit coefficients change considerably: the surface coefficient is −21.69 MeV (as against −26.95 MeV), the surface symmetry coefficient 42.32 MeV (as against 55.62 MeV) and the curvature coefficient is 6.02 MeV (as against 14.77 MeV). The effective Wigner coefficient is −69.47 MeV, markedly larger than the −43.40 MeV of the standard fit,

when a constrained surface Wigner term is included. Once again, values of fit coefficients are significantly affected by inclusion of terms whose effect on the rms deviation is very small.

It might be felt that the equal- $a$  constraint is artificial, as it actually is, there being no obvious argument for such a limitation. A fit without this constraint, but including a surface Wigner term, gives fit coefficients largely unchanged from the standard fit, except for the Wigner coefficient  $\alpha_W$ , which becomes  $-84.02$  MeV (previously  $-43.40$  MeV). This corresponds to a value of  $a = 1.32$ . The surface Wigner coefficient of  $80.69(658)$  MeV corresponds to a value of  $a = 0.70$ . The two  $a$  values differ. The overall improvement in rms deviation, by 52 keV to 1.568 MeV, again fails to justify expansion of the formula, though the sizable change in  $\alpha_W$  once more calls into question the degree to which such global fits can pin down individual effects, like the Wigner term.

#### 5.4. Compact constrained form

In the spirit of the previous subsection, it is possible to extend the unification of terms by allowing a symmetry correction to the volume, surface and curvature terms, while incorporating a Wigner-like  $T(T + a)$  structure, but constraining both the relative symmetry contribution and the Wigner coefficient  $a$  to be the same in all three terms. The compact mass formula so obtained is

$$B = [\alpha_v A + \alpha_s A^{2/3} + \alpha_R A^{1/3}] \left[ 1 + 4\beta_t \frac{T(T + a)}{A^2} \right] + \alpha_C \frac{Z^2}{A^{1/3}} + \alpha_{xC} \frac{Z^{4/3}}{A^{1/3}} + \alpha_p \frac{\delta(N, Z)}{\sqrt{A}} + \alpha_m P + \beta_m P^2 \quad (5.1)$$

and the coefficients are determined by least-squares linear regression on a grid of  $\{\beta_t, a\}$  values followed by a search for the minimum  $D_{\text{rms}}$  on the grid. The resulting best fit has an rms deviation of 1.688 MeV, only 68 keV larger than that of Eq. (3.4), with the coefficients shown in Table 12. The fit is very similar to the ten-term fit, except that the Wigner  $a$  coefficient is somewhat smaller.

An unconstrained fit, with different values allowed for  $a$  and  $\beta_t$  in the volume, surface and curvature terms, produces a slightly reduced rms deviation (by 146 keV), but with rather strange values of some of the coefficients, with very large errors. For example,  $\alpha_W = -179.99(1355)$  MeV, corresponding to  $a = 2.55$ , while the surface Wigner coefficient is  $475.94(5653)$  MeV and  $\alpha_{st} = 93.78(435)$  MeV, corresponding to  $a = 2.54$  and  $\beta_t = -3.94$ . The curvature symmetry coefficient is  $-79.36(901)$  MeV, the curvature Wigner coefficient  $-322.69(5185)$  MeV, corresponding to  $a = 2.03$  and  $\beta_t = -7.96$ . Not every improvement in rms deviation signals a desirable modification of the mass formula!

Table 12

Best fit values (in MeV) of coefficients of the compact semi-empirical mass formula of Eq. (5.1)

| $a$        | $\beta_t$     | $\alpha_v$ | $\alpha_s$ | $\alpha_R$ |
|------------|---------------|------------|------------|------------|
| 0.42       | -1.77         | 16.07(3)   | -23.45(26) | 8.19(40)   |
| $\alpha_C$ | $\alpha_{xC}$ | $\alpha_p$ | $\alpha_m$ | $\beta_m$  |
| -0.757(2)  | 2.13(6)       | 10.00(43)  | -1.82(5)   | 0.128(5)   |



### 5.5. Separate even–even and odd–odd pairing

It was remarked in Section 3.4, in connection with Fig. 7, that even–even mass parabolas lie higher than the interpolated odd-nucleus mass parabolas, while odd–odd mass parabolas lie lower than the interpolated ones. However, in contrast to the usually-adopted form of the pairing term, the offset between the even–even and interpolated-odd mass parabolas is found to be larger than that between the interpolated-odd and odd–odd mass parabolas. The discrepancy could be attributed to residual neutron–proton pairing between the odd nucleons and suggests that the pairing term in the mass formula might be improved by allowing different coefficients for even–even and odd–odd pairing. The pairing term would then take the form

$$\{\alpha_p^{(e)}[1 + \delta(N, Z)] + \alpha_p^{(o)}[1 - \delta(N, Z)]\}\delta(N, Z)/2\sqrt{A}, \quad (5.2)$$

with  $\delta(N, Z)$  defined in Eq. (3.2).

The resulting best fit has almost the same rms deviation as the fit with the standard pairing term (just 3 keV smaller, at 1.618 MeV) and much the same values of the fit coefficients, with the unified  $\alpha_p = 9.87(41)$  MeV being replaced by  $\alpha_p^{(e)} = 11.60(71)$  MeV and  $\alpha_p^{(o)} = 8.17(70)$  MeV. It can hardly be argued that the fit demands different even–even and odd–odd pairing coefficients, but they can be accommodated without any deterioration of the fit.

### 5.6. Binding energy per nucleon

One of the important uses of the semi-empirical mass formula is in establishing the binding energy per nucleon of infinite nuclear matter. The formula is extrapolated to  $A \rightarrow \infty$ , where all terms except the Coulomb term become negligible in comparison with the leading volume term, and the Coulomb term is removed. The binding energy per nucleon of infinite nuclear matter is then identified with the coefficient of the volume term, 16.58(6) MeV in the best fit of Eq. (3.4) or 15.36(3) MeV in the classic BW fit.

Since the quantity of interest for infinite nuclear matter is the binding energy per nucleon, it might appear more logical to fit a semi-empirical formula to the quantity  $B/A$ , rather than to first fit  $B$  and then divide by  $A$ . Such a fit is easily performed, remembering to modify the terms of Eq. (3.4) by dividing each one by  $A$ . The result of a best fit to the BW form of  $B/A$  is compared with the result of a BW fit to  $B$  in Table 13, where it is seen that the corresponding fit coefficients differ quite appreciably between the two fits. However, when the fits are expanded to include all the terms of Eq. (3.4), divided by  $A$  for the second fit, the differences between the fit coefficients, as shown in Table 14, become proportionately much smaller, the prominent exception being the Wigner term. (It is of interest that the rms deviation for the fit to  $B/A$  falls from 137 keV for the BW fit to 42 keV for the full fit, an improvement of a factor of 3.)

The nuclear matter extrapolation is somewhat more robust for the full fit, as compared to the BW fit, though the resulting binding energy per particle of nuclear matter is rather higher than the conventional value.

Table 13

Values (in MeV) of BW coefficients for fits to the binding energy and to the binding energy per nucleon

| Fit   | $\alpha_v$ | $\alpha_s$ | $\alpha_C$ | $\alpha_t$ |
|-------|------------|------------|------------|------------|
| $B$   | 15.36(3)   | −16.42(8)  | −0.691(2)  | −22.53(7)  |
| $B/A$ | 14.14(3)   | −13.68(8)  | −0.585(4)  | −16.56(11) |

Table 14

Best fit values (in MeV) of coefficients of the semi-empirical mass formula of Eq. (3.4) and of the corresponding formula for  $B/A$

| Fit   | $\alpha_v$    | $\alpha_s$ | $\alpha_C$ | $\alpha_t$ | $\alpha_{xC}$ | $\alpha_W$  |
|-------|---------------|------------|------------|------------|---------------|-------------|
| $B$   | 16.58(6)      | −26.95(35) | −0.774(2)  | −31.51(27) | 2.22(5)       | −43.40(222) |
| $B/A$ | 17.12(4)      | −28.08(18) | −0.790(3)  | −33.16(16) | 1.52(4)       | −24.63(47)  |
| Fit   | $\alpha_{st}$ | $\alpha_p$ | $\alpha_R$ | $\alpha_m$ | $\beta_m$     |             |
| $B$   | 55.62(140)    | 9.87(41)   | 14.77(62)  | −1.90(5)   | 0.140(5)      |             |
| $B/A$ | 51.25(48)     | 8.93(12)   | 15.43(22)  | −1.51(9)   | 0.131(15)     |             |

### 5.7. Coulomb radii

The coefficient of the Coulomb term in the mass formula is frequently interpreted in terms of a radius parameter, by regarding the Coulomb term as the self-energy of a uniform spherical charge distribution of radius  $R$  and total charge  $Ze$ . This energy is  $3Z^2e^2/5R$ . The mean squared radius of the charge distribution is  $3R^2/5$ . From the nuclear radius systematics described in Section 2, the rms radius can reasonably be represented as  $\langle r^2 \rangle^{1/2} = r_0 A^{1/3}$ , or the radius of the uniformly charged sphere with the same rms radius as  $R = R_0 A^{1/3}$ , which implies  $R_0 = \sqrt{5/3} r_0$ . Rewriting the Coulomb energy as  $(3e^2/5R_0)(Z^2/A^{1/3})$  allows the extraction of  $R_0$  via  $R_0 = 3e^2/5|\alpha_C|$ . The mass formula fits produce values of  $\alpha_C$  ranging from −0.673 to −0.776 MeV, corresponding to values of  $r_0$  from 0.994 to 0.862 fm, or of  $R_0$  from 1.284 to 1.113 fm. (The conversion factor is  $r_0 = 0.669/|\alpha_C|$  or  $R_0 = 0.864/|\alpha_C|$ , with  $\alpha_C$  in MeV and the radius parameters in fm.)

### 5.8. Weighted fits

All the fits reported above were carried out with the measured binding energies as input, with all data equally weighted. However, nuclear masses are measured to varying degrees of precision, the errors ranging from fractions of an eV for certain nuclides of fundamental importance to almost 2 MeV for the poorest-known nuclide. The mean measurement error for all 2225 nuclei included in the present fits is 42.8 keV, with a standard deviation of 112.5 keV. The error exceeds 100 keV in 229 nuclei and is less than 10 keV in 1017 nuclei. It might be argued that the data should be weighted so as to give greater weight to the more precisely known masses, with very poorly known masses playing a much reduced role. However, the standard procedure of weighting the data with the inverse square of the measurement error would confer overwhelming weight on the few most precisely known masses (43 of them have a quoted error less than 0.1 keV) and would certainly distort the fit. A popular solution is to impose an artificial lower limit, denoted  $\Delta_L$ , on the measurement error. The weight assigned to each data point is then  $1/[\max(\Delta_{\text{exp}}, \Delta_L)]^2$ . The effect of such weighting on the fit is shown in Table 15, which lists the best fit coefficients and rms deviations for several choices of  $\Delta_L$  (where  $\Delta_L = 0$  is the standard inverse-squared-error weighting), as compared with the unweighted fit.

The constrained weighted fits, regardless of the constraint, do not differ very much from the unweighted fit, in the values of the coefficients or in the rms deviation, with the clear exception of the inverse-squared-error weighting, which has an rms deviation greater than 31 MeV and produces a number of coefficients very different from the other fits. As expected, the overwhelming weighting of the few extremely precisely measured masses distorts the fit significantly.

Table 15

Best fit values (in MeV) of coefficients of the semi-empirical mass formula of Eq. (3.4) and rms deviations (in keV), for different weightings. The weighting constraint  $\Delta_L$  (in keV) is defined in the text. The results of the unweighted fit are shown for comparison

|               | Unweighted  | $\Delta_L = 0$ | $\Delta_L = 20$ | $\Delta_L = 50$ | $\Delta_L = 100$ |
|---------------|-------------|----------------|-----------------|-----------------|------------------|
| $\alpha_v$    | 16.58(6)    | 15.00(4)       | 16.57(6)        | 16.57(6)        | 16.55(6)         |
| $\alpha_s$    | −26.95(35)  | −18.24(18)     | −26.42(37)      | −26.58(37)      | −26.57(36)       |
| $\alpha_C$    | −0.774(2)   | −0.577(9)      | −0.767(3)       | −0.769(3)       | −0.769(3)        |
| $\alpha_t$    | −31.51(27)  | −29.33(78)     | −32.01(29)      | −32.00(28)      | −31.79(28)       |
| $\alpha_{xC}$ | 2.22(5)     | −0.43(10)      | 1.89(7)         | 1.98(6)         | 2.07(6)          |
| $\alpha_W$    | −43.40(222) | −37.20(110)    | −47.62(232)     | −47.56(233)     | −45.57(228)      |
| $\alpha_{st}$ | 55.62(140)  | 61.20(231)     | 58.04(160)      | 58.23(153)      | 57.31(147)       |
| $\alpha_p$    | 9.87(41)    | 9.53(2)        | 10.25(43)       | 10.31(43)       | 10.04(42)        |
| $\alpha_R$    | 14.77(62)   | 7.33(14)       | 14.33(62)       | 14.54(63)       | 14.40(63)        |
| $\alpha_m$    | −1.90(5)    | −1.35(3)       | −1.75(5)        | −1.81(5)        | −1.87(5)         |
| $\beta_m$     | 0.140(5)    | 0.730(18)      | 0.123(5)        | 0.131(5)        | 0.137(5)         |
| $D_{rms}$     | 1620.4      | 31659.2        | 1654.8          | 1640.8          | 1629.3           |

Otherwise, any reasonable weighting gives rise to results so similar to those obtained without weighting that there would appear to be no justification for introducing the added complexity of weighted fits.

## 6. Summary and discussion

The classic four-term Bethe–Weizsäcker mass formula (BW) has been supplemented by six additional terms, leading to a reduction in rms deviation from the 2225 measured binding energies of the AME2003 evaluation, from 3536 to 1620 keV, an improvement of more than a factor of two. The six added terms are an exchange Coulomb term, a Wigner term, a surface symmetry term, a pairing term, a curvature term and a coarse-grained global shell correction term. On the basis of their effects on the rms deviation, the most important of the extra terms are the shell correction, surface symmetry and exchange Coulomb terms.

To assess the mutual influence of different terms of the mass formula on one another, fits to the data were performed for all possible selections of terms, from the four-term BW formula to the full ten-term formula, a total of 64 fits. Each fit produced a set of coefficients, with standard errors, and an rms deviation from the data. The systematic pattern of values of fit coefficients and rms deviations was analyzed for correlations between various terms of the formula. In addition, diagonalization of the error matrix for the ten-term fit provided insight into the independent combinations of fit coefficients.

It should be noted that an investigation [16] similar in spirit to the present work, but somewhat more limited in focus, has also established the existence of significant correlations between symmetry, surface symmetry and Wigner terms. It employed the non-linear form of the surface symmetry term and a global shell correction based on the interacting-boson model  $F_{max}$  rather than Casten's  $P$ . This latter choice produces results very similar to, but marginally more successful than, the one adopted here.

The principal conclusions of the present analysis, not necessarily in order of importance, are discussed in turn in the following subsections.

### 6.1. Coulomb energy

The simple single-term Coulomb energy of the BW formula, when complemented by an exchange term, becomes  $\alpha_C \frac{Z^2}{A^{1/3}} [1 + \frac{\alpha_{xC}}{\alpha_C} Z^{-2/3}]$ . The relative contribution of the exchange term depends only on proton number  $Z$  and, with the coefficients of the ten-term fit,  $\alpha_{xC}/\alpha_C = -2.86(7)$ , decreases from 30.1% at  $Z = 20$ , to 16.3% at  $Z = 50$ , to 11.9% at  $Z = 80$  and to 10.3% at  $Z = 100$ . The Coulomb coefficient  $\alpha_C$  goes from  $-0.691(2)$  MeV in the BW fit to  $-0.774(2)$  MeV in the full ten-term fit, but is accompanied in the latter by a compensating exchange contribution. For an extreme heavy nucleus (say  $Z = 100$ ,  $A = 260$ ), the BW estimate of the Coulomb energy is 1082.6 MeV, while the ten-term estimate is  $(1212.7 - 161.2) = 1051.5$  MeV, a less than 3% change, despite the 12% change in  $\alpha_C$ . The combination of direct and exchange Coulomb terms is quite stable, though the coefficients do vary, depending on the terms included in the fit. The direct and exchange Coulomb terms are anti-correlated.

If the Coulomb coefficients  $\alpha_C$  and  $\alpha_{xC}$  are taken from the combined fit to mirror nuclei and Coulomb displacement energies described in Section 3.1, then the relative contribution of the exchange term is only 60% of the results quoted in the previous paragraph. When expressed in terms of direct and exchange contributions to the mirror nucleus energy difference, this still exceeds the single-particle model estimate [17] by a factor of three at  $Z = 20$ .

### 6.2. Wigner term

Half of the fits performed include a Wigner term. Of them, a half also involve a surface symmetry term. In all 16 fits containing a Wigner term but no surface symmetry term, the Wigner coefficient  $\alpha_W$  is positive, which is the wrong sign for a  $T(T + 1)$  or  $T(T + 4)$  type behaviour of the mass formula. The symmetry coefficient  $\alpha_I$  is always negative, so a positive  $\alpha_W$  implies a  $T(T - a)$  form, with  $a > 0$ . However, in all 16 fits containing both a Wigner and a surface symmetry term,  $\alpha_W$  is negative, the right sign for a  $T(T + a)$  behaviour, with  $a > 0$ . The value of  $a$  tends to be somewhat smaller than 1. The Wigner and surface symmetry terms are significantly correlated.

The correlation between Wigner and surface symmetry terms is also evidenced by the rms deviations of the fits. The average contribution of the Wigner term to reduction of the rms deviation is almost 500 keV in the fits with no surface symmetry term, but falls to only 80 keV when a surface symmetry term is included. Similarly, the average improvement in rms deviation due to the surface symmetry term is 725 keV when no Wigner term is present, falling to 350 keV when the Wigner term is included. Both terms play a significant role, but the individual contribution of either term is larger without than with the other. Again, the two terms are clearly correlated.

The values of the fit coefficients  $\alpha_W$  and  $\alpha_{st}$  also show a strong correlation, with a trend coefficient  $R^2 = 0.90$ . The Wigner coefficient is even more strongly correlated with the regular symmetry coefficient  $\alpha_I$ , with  $R^2 = 0.99$ , but the independent combinations of terms obtained by diagonalizing the error matrix show a preference for a mixture of Wigner and surface symmetry terms alone, with some involvement of the curvature term.

An exponential form for the Wigner term, which attains its maximum value at  $N = Z$  and falls away rapidly as  $|N - Z|$  increases, is only marginally favoured over the linear form, by a near-negligible 45 keV improvement in rms deviation. However, with this form, several of the other fit coefficients are significantly changed from their values with a linear term, and also vary

by large amounts as the rate of exponential falloff is modified, with almost no change in rms deviation.

### 6.3. Pairing term

Although its contribution to improving the fit is not large, the pairing term is remarkably stable, both the reduction in rms deviation and the value of the fit coefficient being largely independent of the other terms included in the fit. An improvement of roughly 150 keV in rms deviation is obtained with  $\alpha_p$  of about 10 MeV. There is essentially no correlation between the pairing term and any other term. Although an analysis of mass parabolas for each value of the mass number  $A$  clearly demonstrates that the added binding of even–even nuclei systematically exceeds the loss of binding of odd–odd nuclei, the global fit shows no clear preference for different even–even and odd–odd pairing coefficients.

### 6.4. Shell corrections

The residuals of the fits show characteristic peaks at neutron and proton magic numbers, especially the numbers 2, 8, 20, 28, 50, 82 and 126. (The lower magic numbers tend to get lost in the noise.) The issue of mass corrections arising from the effects of shell closure at the magic numbers has been a long-time focus of microscopic nuclear theory, with some major successes, but there is no known general  $N, Z$  dependence which could form the basis of an additional term in the semi-empirical mass formula. The heuristic approach adopted here, a global parabolic dependence on the valence-nucleon parameter  $P$  of Eq. (3.3), has no pretensions of providing deep theoretical insight, but has proved to be of consistent value in seriously improving the quality of the fits. It reduces the rms deviation by 500 to 770 keV and is to a large extent uncorrelated with other mass formula terms. Proper microscopic shell corrections lead to much greater improvements, achieving even order-of-magnitude reductions in rms deviation [18], but the stable relative contribution of this simple coarse-grained global correction is nonetheless quite impressive.

### 6.5. Surface symmetry term

The surface symmetry term plays an important role in the fit. Aside from its own sizable contribution to reducing the rms deviation, it is solely responsible for ensuring the correct sign of the Wigner coefficient and assists in driving the surface and symmetry coefficients to larger absolute values. It is also rather strongly correlated with the exchange Coulomb term. The standard linear form and the suggested non-linear form give rise to very similar rms deviations, but do produce rather different values of some of the other coefficients. (The Wigner coefficient  $\alpha_W$  changes from  $-45$  to  $-10$  MeV in going from the linear to the non-linear form of the surface symmetry term.) The global fit alone makes no compelling case for either the linear or the non-linear form of the surface symmetry term. Each form has its arguments in favour. But the sensitivity of some other coefficients to the choice of form raises questions of principle about the degree to which the semi-empirical approach can determine the values of the fit parameters.

There is little doubt that an acceptable semi-empirical mass formula must include a surface symmetry term, but its coefficient is not very strongly constrained.

## 6.6. Symmetry term

The symmetry coefficient  $\alpha_t$  takes the value  $-22.5$  MeV in the BW fit,  $-27.3$  MeV in the extended BW fit (including a surface symmetry term) and reaches  $-31.5$  MeV in the full ten-term fit. (In a few fits, among them the fit to binding energy per nucleon, it gets to  $-33$  MeV.) It is quite strongly correlated with the exchange Coulomb term. The symmetry term is of considerable current interest in connection with the equation of state of nuclear and neutron matter and its implications for the structure of neutron stars. Analysis of heavy-ion experiments appears to favour the larger values of  $|\alpha_t|$ , and perhaps even the largest values of those quoted.

## 6.7. The best fits

Of the 64 fits studied here, five have rms deviations below 1825 keV, i.e. within 200 keV of the lowest rms deviation. Taking into account in addition the refinements discussed in Section 5, there are nine fits with rms deviations between 1568 and 1754 keV. They are (with their rms deviations in brackets): The fit including a surface Wigner term (1568 keV); the fit including an exponential Wigner term (1576 keV); the fit with distinct even–even and odd–odd pairing (1618 keV); the fit with common  $T(T+a)$  dependence for volume and surface symmetry terms (1620 keV); the standard fit (1620 keV); the fit with common volume and surface dependence on neutron excess (1632 keV); the compact fit with a common  $T(T+a)$  dependence for volume, surface and curvature terms (1689 keV); the fit with a non-linear surface symmetry term (1743 keV); and the fit containing no Wigner term (1754 keV). (Here the term “fit” refers to a full 10-term formula, as in Eq. (3.4), modified as indicated when necessary.) A rather good indication of the extent to which individual coefficients can be determined by such semi-empirical fits is obtained by noting the range of values of each coefficient in these nine most-favoured fits, as detailed in Table 16.

Inspection of the table reveals that  $\alpha_v$ ,  $\alpha_C$ ,  $\alpha_p$ ,  $\alpha_m$  and  $\beta_m$  are reasonably well determined by semi-empirical fits, that  $\alpha_s$ ,  $\alpha_t$  and  $\alpha_{xC}$  have significant flexibility, and that  $\alpha_{st}$ ,  $\alpha_R$  and especially  $\alpha_W$  are hardly determined at all. The question poses itself whether it might not be preferable to try to fix some parameters by more “local” filtered arguments and then to use the global mass fits to specify the remaining parameters. Examples might be the use of mirror nucleus energy differences and Coulomb displacement energies to set  $\alpha_C$  and  $\alpha_{xC}$ , or the use of mass parabolas to pin down  $\alpha_p$  (or possibly separate  $\alpha_p$ ’s for even–even and odd–odd nuclei).

Table 16

Range of values (in MeV) of coefficients of the semi-empirical mass formula for the nine fits with lowest rms deviation. The column labeled “min” shows the lowest (in absolute value) value attained by each coefficient in the nine favoured fits, the column labeled “max” the highest value

| Coefficient   | min    | max    | Coefficient | min   | max   |
|---------------|--------|--------|-------------|-------|-------|
| $\alpha_v$    | 15.8   | 16.6   | $\alpha_W$  | –10.1 | –84.0 |
| $\alpha_s$    | –19.2  | –27.0  | $a$         | 0.14  | 1.32  |
| $\alpha_C$    | –0.745 | –0.777 | $\alpha_p$  | 9.7   | 10.5  |
| $\alpha_t$    | –26.8  | –34.9  | $\alpha_R$  | –6.9  | 14.8  |
| $\alpha_{xC}$ | 1.5    | 2.6    | $\alpha_m$  | –1.80 | –1.90 |
| $\alpha_{st}$ | 29.9   | 57.5   | $\beta_m$   | 0.125 | 0.140 |

## 6.8. Summing up

It may fairly be concluded that the classic Bethe–von Weizsäcker semi-empirical mass formula can be usefully supplemented by additional terms, of which the most important is probably the surface symmetry term. This term is solely responsible for the correct sign of the Wigner term and largely responsible for the large magnitude of the symmetry coefficient favoured by recent heavy-ion experiments. In addition, even a coarse, global shell correction can significantly reduce the rms deviation of the fit to the measured masses. A pairing term produces a small but robust improvement in the fit, essentially independent of which other terms are included in an extended mass formula. Similarly, an exchange Coulomb term can have an important effect, but this depends rather more on just which other terms are present. The Wigner term, and a curvature term, do not very strongly affect the quality of the fit once a surface symmetry term is included but can have a noticeable influence on determining the fit parameters.

There are a number of possible combinations of added terms which produce almost equally good fits to the known mass data. In addition, alternative forms of some of the terms lead to almost equivalent fits, but with rather different values of the fit coefficients, including some of those of the classic four-term formula. Some constrained fits, such as uniform  $T(T + a)$  behaviour or uniform  $(N - Z)^2/A^2$  dependence, do as well as, or even better than, some of best of the free fits, again with some variability of fit parameters.

Within the framework of global mass fits, the volume, Coulomb and pairing coefficients are reasonably well determined, as are the coefficients of a coarse-grained shell correction. The surface, symmetry and exchange Coulomb coefficients have a certain amount of play, while the curvature and surface symmetry coefficients are rather poorly established and the Wigner coefficient is almost indeterminate.

There are significant correlations among the different terms of the mass formula. These, together with the flexibility of form of some of the terms, make the standard errors of the linear least-squares fits poor indicators of the true uncertainty in the values of the parameters of the mass formula.

The parameters of the semi-empirical mass formula can probably be quite reliably estimated as lying in roughly the following ranges of values:  $\alpha_v \sim 16$  MeV;  $\alpha_s \sim -20 \rightarrow -25$  MeV;  $\alpha_C \sim -0.77$  MeV;  $\alpha_t \sim -30$  MeV;  $\alpha_{xC} \sim 2$  MeV;  $\alpha_{st} \sim 30 \rightarrow 55$  MeV;  $\alpha_W \sim -20 \rightarrow -70$  MeV;  $\alpha_p \sim 10$  MeV;  $\alpha_R \sim -7 \rightarrow 14$  MeV;  $\alpha_m \sim -1.8$  MeV;  $\beta_m \sim 0.13$  MeV.

## Acknowledgements

The support of the Joseph and Bessie Feinberg Foundation is gratefully acknowledged.

## References

- [1] H.A. Bethe, R.F. Bacher, *Rev. Mod. Phys.* 8 (1936) 82–229.
- [2] J. Duflo, A.P. Zuker, *Phys. Rev. C* 52 (1995) R23–R27.
- [3] G. Royer, C. Gautier, *Phys. Rev. C* 73 (2006) 067302.
- [4] G. Audi, A.H. Wapstra, C. Thibault, *Nucl. Phys. A* 729 (2003) 337–676.
- [5] I. Angeli, *At. Data Nucl. Data Tables* 87 (2004) 185–206.
- [6] A. Bohr, B.R. Mottelson, *Nuclear Structure*, vol. 1, Benjamin, New York, 1969, pp. 220ff.
- [7] I. Talmi, *Simple Models of Complex Nuclei*, Harwood Academic, Switzerland, 1993.
- [8] W.D. Myers, W.J. Swiatecki, *Ann. Phys. (N.Y.)* 55 (1969) 395–505.
- [9] M.S. Antony, A. Pape, J. Britz, *At. Data Nucl. Data Tables* 66 (1997) 1–63.

- [10] A. de Shalit, H. Feshbach, *Theoretical Nuclear Physics, Nuclear Structure*, vol. 1, John Wiley & Sons, New York, 1974, p. 140.
- [11] E.P. Wigner, *Phys. Rev.* 51 (1937) 106–119;  
E.P. Wigner, *Phys. Rev.* 51 (1937) 947–958.
- [12] W.D. Myers, W.J. Swiatecki, *Nucl. Phys.* 81 (1966) 1–60.
- [13] P. Danielewicz, *Nucl. Phys. A* 727 (2003) 233–268.
- [14] R.F. Casten, *Nuclear Structure from a Simple Perspective*, Oxford Univ. Press, Oxford, 2000.
- [15] I. Angeli, *Europhys. Lett.* 16 (1991) 429–432.
- [16] A.E.L. Dieperink, P. Van Isacker, *Eur. Phys. J. A* 32 (2007) 11–18.
- [17] S. Shlomo, *Rep. Prog. Phys.* 41 (1978) 957–1026.
- [18] J.G. Hirsch, A. Frank, J. Barea, P. Van Isacker, V. Velázquez, *Eur. Phys. J. A* 25 (2005) 75–78.